

ASBESTOS REINSPECTION REPORT

LOCATION:

**JESSE BROWN VA MEDICAL CENTER
820 S DAMEN AVE.
CHICAGO, IL 60612**



**PROJECT NO.: 537-19-122
TASK ORDER: 36C25219F0327**



**INSPECTION DATES:
SEPTEMBER 3-6, 2019**

PRIME CONTRACTOR:

**CASTLE HILL ASSOCIATES, LLC
218-A MECHANIC ST.
WATERVILLE, OH 43566
567-952-0212**



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1.0 EXECUTIVE SUMMARY

From September 3-6, 2019, Castle Hill Associates, LLC was contracted by the US Department of Veterans Affairs (VA) to perform asbestos re-inspections of buildings 1A, 1B, 11A, and 11B on the site of the Jesse Brown VA Medical Center in Chicago, IL. The survey was performed as part of the VA Project Number 537-19-122 under Task Order 36C25219F0327. The Performance Work Statement (PWS) issued by the VA defined the scope of work for the asbestos survey.

The inspection was conducted by Brett Brumbaugh and Sam Ansara. The project manager was Mr. Brumbaugh who has been conducting asbestos surveys, sampling, and consulting work since 1987. He is a State of Illinois certified Asbestos Inspector. Sam Ansara has been conducting asbestos surveys, sampling, and consulting work since 1989. There was a previous asbestos reinspection and sampling conducted at the facility most recently in 2007 and those reports were reviewed and utilized by Castle Hill prior to this inspection. There were no limitations or exceptions to the PWS for the surveys conducted.

All accessible functional areas in the buildings were surveyed. All previous asbestos survey information and sampling conducted at the facilities were evaluated and incorporated into this survey. All confirmed asbestos-containing materials were verified. This report can be incorporated into the facility's management plan for asbestos-containing materials.

2.0 STATE AND LOCAL REGULATIONS

The State of Illinois asbestos inspection and licensing regulations are similar to the Federal regulations. State asbestos regulations include requirements for licensing and certification of personnel who will perform asbestos-related work activities. All personnel who engage in asbestos-related activities must comply with the state licensing/certification requirements as well as with all Federal requirements.

3.0 ANALYTICAL METHODS AND LIMITATIONS

3.1 Asbestos Sampling Protocol

EPA and OSHA define an ACM as any material which contains greater than 1 percent asbestos. The ACM inspection and bulk sampling were performed in accordance with the methods outlined in the EPA guidance document titled *Guidance for Controlling Asbestos-Containing Materials in Buildings* (Document No. 560/5-85/024). In addition, bulk sampling of asbestos was performed in accordance with 40 CFR Part 763, AHERA requirements for number of samples and types of ACM to be sampled. According to these requirements, materials are classified as either surfacing (e.g., decorative plaster, spray-applied fireproofing), thermal system insulation (e.g., pipe insulation, pipe fitting insulation, boiler insulation), or miscellaneous materials (e.g., floor tile, ceiling tile, plaster, wallboard, mastics).

The number of samples collected from each material varies based on the classification of the material and increases as the potential for a non-uniform mixture of asbestos in the material increases. Typically, regulations require a minimum number of samples to be collected from the different classifications of suspect ACM.

All bulk samples were analyzed utilizing current EPA methods for PLM (Polarized Light Microscopy) by EMSL Analytical in Plymouth, MI who accredited by the NIST, NVLAP program (National Institute of Standards and Technology, National Voluntary Laboratory Accreditation Program) and also an American Industrial Hygiene Association (AIHA) accredited laboratory.

4.0 SUMMARY OF FINDINGS

4.1 Asbestos-Containing Building Materials

The following is a summary of the asbestos containing materials, locations, quantities of materials in each inspected building for this project:

Building 1A – Damen Pavillion

Asbestos Containing Materials	Locations (see Appendix G for exact locations)	Quantity
Pipe insulation	Inaccessible wall and ceiling chases (above/behind hard plaster); Mech. Rm. P400; Basement hall by B403	460 lf known Unknown LF in chases
Mud fittings on fiberglass insulated pipe	Hall by 1510-15; 12 th floor elev. mech. room	46 fittings known Unknown # in chases
12" vinyl floor tile (all colors) and/or black mastic	1 st through 10 th floors	76,588 sf
9" vinyl floor tile and black mastic	9 th floor	580 sf
Transite cement sheets behind radiators	7 th through 10 th floors	2,760 sf

Building 1B – Damen Pavillion

Asbestos Containing Materials	Locations (see Appendix G for exact locations)	Quantity
Pipe insulation	Inaccessible wall and ceiling chases (above/behind hard plaster)	420 lf known Unknown LF in chases
Mud fittings on fiberglass insulated pipe	Room 10503; Inaccessible wall and ceiling chases (above/behind hard plaster)	8 fittings known Unknown # in chases
12" vinyl floor tile (all colors) and/or black mastic	1 st , 2 nd , 3 rd , 4 th , 6 th , 8 th , and 10 th floors	38,340 sf

Building 11A – Taylor Building

Asbestos Containing Materials	Locations (see Appendix G for exact locations)	Quantity
Pipe insulation	Inaccessible wall and ceiling chases (above/behind hard plaster)	Unknown LF in chases
Mud fittings on fiberglass insulated pipe	Inaccessible wall and ceiling chases (above/behind hard plaster)	8 fittings known Unknown # in chases
12" vinyl floor tile (all colors) and/or black mastic	6 th floor	3,640 sf
Laboratory Fume Hoods (Transite)	6219, 6220, 6221, 6256, 6260, 6261, 6273, 6281, 6283, 6285, 7203, 7211, 7218, 7235, 7237, 7239, 7241, 7275, 7281, 7282, 7283, 7287, 7288, 7290, 7288, 7292, 7293, 7298, 7299	2,240 sf
Lab Table Tops, Sinks, and Drying Racks (Ebony Transite)	6219, 6220, 6221, 6256, 6260, 6261, 6273, 6281, 6283, 6285, 7203, 7211, 7218, 7235, 7237, 7239, 7241, 7275, 7281, 7282, 7283, 7287, 7288, 7290, 7288, 7292, 7293, 7298, 7299	5,670 sf 78 sinks
Transite sheets in stairwells	Stairwell 11 to 8 th floor penthouse	70 sf

Building 11B – Ogden Building

Asbestos Containing Materials	Locations (see Appendix G for exact locations on each floor)	Quantity
Contaminated soil in crawl space	Crawl spaces	2,376 sf
12" vinyl floor tile (all colors) and black mastic	3 rd floor	3,318 sf
Pipe insulation	Vertical pipe chases	Unknown quantity
Mud fittings on fiberglass insulated pipe	Vertical pipe chases	Unknown quantity
Transite sheets in stairwells	Stairwell 7 (basement and 7 th floors)	320 sf

5.0 Explanation of Assessment of Asbestos Containing Materials

Re-inspection protocol is based on the Asbestos Hazard Emergency Response Act (AHERA), Environmental Protection Act's (EPA's) revised Asbestos Model Accreditation Plan (MAP) mandated by the Asbestos School Hazard Abatement Reauthorization Act (ASHARA), and the Department of Veterans Affairs (VA) Directive 7700. The tabular listing included in this report contains Building Number, Floor, Room Number, Material Category (i.e., TSI, SM or MM), Material Type, quantity, a numerical priority ranking, condition, potential for damage, potential for human exposure, drawing number, hazard ranking, recommended response action and associated cost estimate for implementation of those response actions.

The material assessment categories of the ACM identified in the building is formatted in the following order for the summary tables enclosed in this report:

- Material Category - Surfacing Material (SUR); Thermal System Insulation (TSI); Miscellaneous (MM)
- Material Type - Friable (F); Non-Friable (NF)
- Condition of Material - Good; Fair; Poor
- Potential for Damage - Low; Medium; High
- Physical Assessment - Low; Medium; High
- Recommended Response Action - 1 through 7

Material category is the type of material such as surfacing material (SUR), thermal system insulation (TSI), and miscellaneous (MM)

- **Surfacing Material (SUR)** - Surfacing ACM means surfacing material that is ACM. Surfacing material means material in a school building that is sprayed-on, troweled-on, or otherwise applied to surfaces, such as acoustical plaster on ceilings and fireproofing materials on structural members, or other materials on surfaces for acoustical, fireproofing, or other purposes.

- **Thermal System Insulation (TSI)** - Thermal system insulation ACM means thermal system insulation that is ACM. Thermal system insulation means material in a school building applied to pipes, fittings, boilers, breeching, tanks, ducts, or other interior structural components to prevent heat loss or gain, or water condensation, or for other purposes.

- **Miscellaneous (MM)** - Miscellaneous ACM means miscellaneous material that is ACM. Miscellaneous material means interior building material on structural components, structural members or fixtures, such as floor and ceiling tiles, and does not include surfacing material or thermal system insulation.

Material type refers to the hardness of a material to be crumbled, pulverized, or reduced to powder by hand pressure as friable (F), and includes previously non-friable (NF) material after such previously non-friable material becomes damaged to the extent that when dry it may be crumbled, pulverized, or reduced to powder by hand pressure.

- Condition of Material - Good; Fair; Poor
- Potential for Damage - Low; Medium; High
- Physical Assessment - Low; Medium; High

Seven Categories define in the AHERA regulation one of which must be assigned to each friable surfacing and miscellaneous ACM and each asbestos-containing TSI during the inspection/re-inspection.

1. Damaged or significantly damaged TSI ACM - Damaged or significantly damaged thermal system insulation ACM means thermal system insulation ACM on pipes, boilers, tanks, ducts, and other thermal system insulation equipment where the insulation has lost its structural integrity, or its covering, in whole or in part, is crushed, water-stained, gouged, punctured, missing, or not intact such that it is not able to contain fibers. Damage may be further illustrated by occasional punctures, gouges or other signs of physical injury to ACM; occasional water damage on the protective coverings/jackets; or exposed ACM ends or joints. Asbestos debris originating from the ACM in question may also indicate damage.
2. Damaged friable surfacing ACM - Damaged friable surfacing ACM means friable surfacing ACM which has deteriorated or sustained physical injury such that the internal structure (cohesion) of the material is inadequate or which has delaminated such that its bond to the substrate (adhesion) is inadequate, or which, for any other reason, lacks fiber cohesion or adhesion qualities. Such damage or deterioration may be illustrated by the separation of ACM into layers; separation of ACM from the substrate; flaking, blistering, or crumbling of the ACM surface; water damage; significant or repeated water stains, scrapes, gouges, mars or other signs of physical injury on the ACM. Asbestos debris originating from the ACM in question may also indicate damage.
3. Significantly damaged friable surfacing ACM - Significantly damaged friable surfacing ACM means damaged friable surfacing ACM in a functional space where the damage is extensive and severe.
4. Damaged or significantly damaged friable miscellaneous ACM - Damaged friable miscellaneous ACM means friable miscellaneous ACM which has deteriorated or sustained physical injury such that the internal structure (cohesion) of the material is inadequate or, if applicable, which has delaminated such that its bond to the substrate (adhesion) is inadequate or which for any other reason lacks fiber cohesion or adhesion qualities. Such damage or deterioration may be illustrated by the separation of ACM into layers; separation of ACM from the substrate; flaking, blistering, or crumbling of the ACM surface; water damage; significant or repeated water stains, scrapes, gouges, mars or other signs of physical injury on the ACM. Asbestos debris originating from the ACM in question may also indicate damage.
5. ACM with Potential for Damage - Potential damage means circumstances in which:
 - (1) Friable ACM is in an area regularly used by building occupants, including maintenance personnel, in the course of their normal activities.
 - (2) There are indications that there is a reasonable likelihood that the material or its covering will become damaged, deteriorated, or delaminated due to factors such as changes in building use, changes in operations and maintenance practices, changes in occupancy, or recurrent damage.
6. ACM with potential for significant damaged - Potential significant damage means circumstances in which:
 - (1) Friable ACM is in an area regularly used by building occupants, including maintenance personnel, in the course of their normal activities.
 - (2) There are indications that there is a reasonable likelihood that the material or its covering will become significantly damaged, deteriorated, or delaminated due to factors such as changes in building use, changes in operations and maintenance practices, changes in occupancy, or recurrent damage.
7. Any remaining friable ACM or Friable Suspected ACM

6.0 EXPLANATION OF REMEDIATION COST ESTIMATES

All remediation cost (removal and repair) estimates are based upon RS Means data. Since RS Means does not account for work in VA Medical Centers, all cost estimates are adjusted to reflect the VA specific requirements for abatement. In addition, the cost estimates reflect the quantity and location of the asbestos materials. For example, abatement work in crawl spaces is expected to be much costlier than work in more accessible areas of building. Also, the costs for each material in each building are intended to be for “stand alone” projects and the combination of areas for removal will reduce the costs due to an economy of scale.

7.0 CONCLUSION/RECOMMENDATIONS

An operations and maintenance (O&M) program has been developed by the VAMC specifically for the positively identified ACBM in the facilities. Any renovation/demolition work which may impact these positive materials should be conducted in accordance with the O&M program and all applicable Federal, state, and local regulations by appropriately trained and VA-approved personnel and contractors. Since demolition was not performed as part of the inspection work at the facilities, certain areas such as pipe chases and areas above fixed permanent ceilings, fire-door interiors, and vault/safe wall and door linings were not inspected or tested. Prior to renovation and alteration work which could impact these areas, Castle Hill recommends that exploratory demolition be performed and a qualified inspector be utilized to examine the areas and collect samples as needed. Castle Hill recommends sampling of roofing materials and electrical components (not specifically examined as part of this survey) if renovation activities are planned which may disturb these materials.

Respectfully Submitted:



Brett L. Brumbaugh, REPA, CESCO
State of Illinois Asbestos Inspector
License Number 100-09972

TABLE 1 LIST OF ACRONYMS

ACBM	Asbestos Containing Building Material
ACM	Asbestos Containing Material
ASHERA	Asbestos Hazard Emergency Response Act
AIHA	American Industrial Hygiene Association
CAA	Clean Air Act
CFR	Code of Federal Regulations
EPA	United States Environmental Protection Agency
HazCom	Hazard Communication
HEPA	High Efficiency Particulate Air Filter
HVAC	Heating, Ventilation, and Air Conditioning
MISC	Miscellaneous Asbestos Containing Material
NESHAP	National Emission Standards for Hazardous Air Pollutants
NIOSH	National Institute for Occupational Safety and Health
NIST	National Institute of Standards and Technology
NOB	Non-friable Organically Bound
NVLAP	National Voluntary Laboratory Accreditation Program
OSHA	Occupational Safety and Health Administration
PAT	AIHA Proficiency Analytical Testing Program
PCM	Phase Contrast Microscopy
PEL	Permissible Exposure Limit
PLM	Polarized Light Microscopy
TEM	Transmission Electron Microscopy
TSCA	Toxic Substances Control Act
TSI	Thermal Systems Insulation
Castle Hill	Castle Hill Associates, LLC
VA	United States Department of Veterans Affairs
VAMC	VA Medical Center
XRD	X-Ray Diffraction

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TABLE 2
ASBESTOS REMOVAL PRIORITY PROTOCOLS

HAZARD RANK	REMOVAL PRIORITY	AHERA CATEGORY	RESPONSE ACTION RECOMMENDATION
7	1	Significantly Damaged	Evacuate or isolate the area. Remove or repair if feasible. O&M for all friable ACM.
6	2	Damaged with potential for significant damage	Evacuate or isolate the area. Remove, encapsulate, or repair if feasible. O&M for all friable ACM.
5	3	Damaged with potential for damage	Remove, encapsulate, or repair. O&M for all friable ACM.
4	4	Damaged	Remove, encapsulate, or repair. O&M for all friable ACM.
3	5	Potential for significant damage	Take steps to reduce potential for damage. O&M for all friable ACM.
2	6	Potential for damage	O&M for all friable ACM.
1	7	No problem	O&M for all friable ACM.

APPENDIX A

ASBESTOS BULK SAMPLING DATA

JESSE BROWN VA MEDICAL CENTER – CHICAGO, IL
ASBESTOS REINSPECTION REPORT
SEPTEMBER 2019

ASBESTOS BULK SAMPLING INFORMATION

Facility: **Jesse Brown VAMC - Chicago, IL**
Buildings: **Buildings 1A, 1B, 11A, and 11B**
Sampling Dates: **September 3-4, 2019**
Analysis Dates: **September 14-16, 2019**
Laboratory: **EMSL Analytical, Inc. – Cinnaminson, NJ**
Accreditation: **NVLAP Asbestos Bulk Accredited Laboratory # 101048-0**
Sampler: **Brett Brumbaugh**
Accreditation: **State of Illinois – Asbestos Inspector #100-09972**

PLM Laboratory Analysis Result

BLDG. #	SAMPLE NUMBER	MATERIAL	SAMPLE LOCATION	LABORATORY ANALYSIS RESULT
11B	1	Spray-on fireproofing	Basement – B301	No Asbestos Detected
11B	2	Spray-on fireproofing	Basement – B301	No Asbestos Detected
11B	3	Spray-on fireproofing	Basement – B301	No Asbestos Detected
11B	4	Transite wall paneling	Basement – Stairwell 7	15% Chrysotile Asbestos
1A	5	Spray-on fireproofing	Elev. Mech. Room 12 th Fl.	No Asbestos Detected
1A	6	Spray-on fireproofing	Elev. Mech. Room 12 th Fl.	No Asbestos Detected
1A	7	Spray-on fireproofing	Elev. Mech. Room 12 th Fl.	No Asbestos Detected
11B	8	Spray-on fireproofing	8 th floor penthouse	No Asbestos Detected
11B	9	Spray-on fireproofing	8 th floor penthouse	No Asbestos Detected
11B	10	Spray-on fireproofing	8 th floor penthouse	No Asbestos Detected
11A	11	Spray-on fireproofing	8 th floor penthouse	No Asbestos Detected
11A	12	Spray-on fireproofing	8 th floor penthouse	No Asbestos Detected
11A	13	Spray-on fireproofing	8 th floor penthouse	No Asbestos Detected
11A	14	Tan duct seam mastic	8 th floor penthouse	No Asbestos Detected
11A	15	Tan duct seam mastic	8 th floor penthouse	No Asbestos Detected
11A	16	Spray-on fireproofing	8 th floor penthouse	No Asbestos Detected
11A	17	Spray-on fireproofing	8 th floor penthouse	No Asbestos Detected
11A	18	Spray-on fireproofing	8 th floor penthouse	No Asbestos Detected
11A	19a	12" white/blue/green floor tile	Room 7221	No Asbestos Detected
11A	19b	Black mastic	"	No Asbestos Detected
11A	20a	12" white/blue/green floor tile	Hall by Room 6233	No Asbestos Detected
11A	20b	Yellow mastic	"	No Asbestos Detected
11A	21a	12" tan floor tile	Hall by Room 6239	3% Chrysotile Asbestos
11A	21b	Black mastic	"	5% Chrysotile Asbestos
11A	22a	12" tan floor tile	Hall by Room 6249	3% Chrysotile Asbestos
11A	22b	Black mastic	"	5% Chrysotile Asbestos
11A	23a	12" white/blue/green floor tile	Hall by Room 4203	No Asbestos Detected
11A	23b	Yellow mastic	"	No Asbestos Detected
11A	24a	12" gray floor tile	Hall by Room 4289	No Asbestos Detected
11A	24b	Yellow mastic	"	No Asbestos Detected
11A	25a	12" beige/brown floor tile	Hall by room 3260	No Asbestos Detected
11A	25b	Yellow mastic	"	No Asbestos Detected
11A	26a	12" beige/brown floor tile	Hall by Room 3229a	No Asbestos Detected
11A	26b	Yellow mastic	"	No Asbestos Detected
11A	27a	12" Gray/white floor tile	Room 2214	No Asbestos Detected
11A	27b	Yellow mastic	"	No Asbestos Detected
11A	28a	12" White floor tile	Hall by Room 2277	No Asbestos Detected
11A	28b	Yellow mastic	"	No Asbestos Detected
11A	29	12" tan/blue/green floor tile	Room 2280	No Asbestos Detected
11A	30	Spray-on fireproofing on column	Mech. room 1280	No Asbestos Detected

**JESSE BROWN VA MEDICAL CENTER – CHICAGO, IL
ASBESTOS REINSPECTION REPORT
SEPTEMBER 2019**

ASBESTOS BULK SAMPLING INFORMATION

Facility: **Jesse Brown VAMC - Chicago, IL**
 Buildings: **Buildings 1A, 1B, 11A, and 11B**
 Sampling Dates: **September 3-4, 2019**
 Analysis Dates: **September 14-16, 2019**
 Laboratory: **EMSL Analytical, Inc. – Cinnaminson, NJ**
 Accreditation: **NVLAP Asbestos Bulk Accredited Laboratory # 101048-0**
 Sampler: **Brett Brumbaugh**
 Accreditation: **State of Illinois – Asbestos Inspector #100-09972**

PLM Laboratory Analysis Result

BLDG. #	SAMPLE NUMBER	MATERIAL	SAMPLE LOCATION	LABORATORY ANALYSIS RESULT
11A	31	Spray-on fireproofing on column	Mech. room 1280	No Asbestos Detected
11A	32	Spray-on fireproofing on column	Mech. room 1280	No Asbestos Detected
1A	40a	12" tan floor tile	Room 10503	No Asbestos Detected
1A	40b	Black mastic	"	4% Chrysotile Asbestos
1A	41a	12" tan floor tile	Hall by 10440	No Asbestos Detected
1A	41b	Black mastic	"	3% Chrysotile Asbestos
1B	42a	12" light tan floor tile	Hall by 10503	2% Chrysotile Asbestos
1B	42b	Black mastic	"	5% Chrysotile Asbestos
1B	43a	12" light tan floor tile	Hall by 10520	3% Chrysotile Asbestos
1B	43b	Black mastic	"	4% Chrysotile Asbestos
1A	44a	12" green floor tile	Hall by 9463A	No Asbestos Detected
1A	44b	Black mastic	"	2% Chrysotile Asbestos
1A	45a	12" tan floor tile	Hall by 9435	No Asbestos Detected
1A	45b	Black mastic	"	No Asbestos Detected
1A	46a	9" tan floor tile	Room 9495	3% Chrysotile Asbestos
1A	46b	Black mastic	"	4% Chrysotile Asbestos
1B	47a	12" tan/black floor tile	Room 8560	7% Chrysotile Asbestos
1B	47b	Black mastic	"	5% Chrysotile Asbestos
1A	48a	12" tan/black floor tile	Room 8425	No Asbestos Detected
1A	48b	12" tan/black floor tile	"	3% Chrysotile Asbestos
1A	49a	12" gray floor tile	Room 7406	4% Chrysotile Asbestos
1A	49b	Black mastic	"	3% Chrysotile Asbestos
1A	50a	12" tan/brown floor tile	Room 7498	No Asbestos Detected
1A	50b	Black mastic	"	3% Chrysotile Asbestos
1A	51a	12" tan/blue/green floor tile	Hall by Rom 3467	No Asbestos Detected
1A	51b	Yellow mastic	"	No Asbestos Detected
1B	52a	12" tan/black floor tile	Hall by 2580	No Asbestos Detected
1B	52b	Black mastic	"	3% Chrysotile Asbestos
11B	53a	12" blue floor tile	Elevator lobby	No Asbestos Detected
11B	53b	Black mastic	"	No Asbestos Detected
11B	54a	12" tan floor tile	Elevator lobby	No Asbestos Detected
11B	54b	Black and yellow mastic	"	No Asbestos Detected
11B	55a	12" gray floor tile (bottom layer)	Hall by Room 4301	No Asbestos Detected
11B	55b	Black mastic (bottom layer)	"	No Asbestos Detected
11B	55c	12" brown wood pattern floor tile	Hall by Room 4301	No Asbestos Detected
11B	55d	Yellow mastic	"	No Asbestos Detected
11B	56a	12" tan/brown floor tile	Room 3375	No Asbestos Detected
11B	56b	Black mastic	"	2% Chrysotile Asbestos
11B	57a	12" olive floor tile	Room 3303	No Asbestos Detected
11B	57b	Black mastic	"	No Asbestos Detected

- EPA Definition of an asbestos containing material: Greater than 1% asbestos



JESSE BROWN VA MEDICAL CENTER – CHICAGO, IL
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ASBESTOS IN SOILS BULK SAMPLING INFORMATION

Facility: **Jesse Brown VAMC - Chicago, IL**
Buildings: **Buildings 11A and 11B**
Sampling Dates: **September 4, 2019**
Analysis Dates: **September 11, 2019**
Laboratory: **EMSL Analytical, Inc. – Cinnaminson, NJ**
Accreditation: **NVLAP Asbestos Bulk Accredited Laboratory # 101048-0**
Sampler: **Brett Brumbaugh**
Accreditation: **State of Illinois – Asbestos Inspector #100-09972**

PLM Qualitative Laboratory Analysis Result (Asbestos in Soils)

SAMPLE NUMBER	MATERIAL	SAMPLE LOCATION	LABORATORY ANALYSIS RESULT
33	Loose soil in crawl space	Crawl space under 11A	No Asbestos Detected
34	Loose soil in crawl space	Crawl space under 11A	No Asbestos Detected
35	Loose soil in crawl space	Crawl space under 11A	No Asbestos Detected
36	Loose soil in crawl space	Crawl space under 11B	Chrysotile Asbestos Present
37	Loose soil in crawl space	Crawl space under 11B	No Asbestos Detected
38	Loose soil in crawl space	Crawl space under 11B	Chrysotile Asbestos Present
39	Loose soil in crawl space	Crawl space under 11B	No Asbestos Detected
58	Loose soil in crawl space	Crawl space under 11B	Chrysotile Asbestos Present

OSHA Definition of an asbestos containing soil: Any percentage of asbestos present

APPENDIX B

Asbestos Quantities, Assessments, & Remediation Cost Estimates

**Jesse Brown VAMC - Chicago, IL
Building 1A
Asbestos Assessments and Cost Estimates
September 2019**

Asbestos Material	Locations	Quantity	AHERA Category	Friability %Damage	AHERA Assessment	Recommendation	Removal Priority	Repair Cost Estimate	Removal Cost Estimate
Pipe insulation	Inaccessible wall and ceiling chases (above/behind hard plaster); Mech. Rm. P400; Basement hall by B403	14220 lf	TSI	Friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$497,000
Mud fittings on fiberglass insulated pipe	Hall by 1510-15; 12th floor elev. mech. room	1330 ea	TSI	Friable 0%	1	O&M; Remove at renovation or demolition	1	No damage	\$99,750
12" vinyl floor tile (all colors) and/or black mastic	1st through 10th floors	76,588 sf	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$459,000
9" vinyl floor tile and black mastic	9th floor	580 sf	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$6,970
Transite cement sheets behind radiators	7th through 10th floors	2,760 sf	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$69,000
TOTALS								0	\$1,131,720

**Jesse Brown VAMC - Chicago, IL
Building 1B
Asbestos Assessments and Cost Estimates
September 2019**

Asbestos Material	Locations	Quantity	AHERA Category	Friability %Damage	AHERA Assessment	Recommendation	Removal Priority	Repair Cost Estimate	Removal Cost Estimate
Pipe insulation	Inaccessible wall and ceiling chases (above/behind hard plaster); Mech. Rm. P400; Basement hall by B403	11,260 sf	TSI	Friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$480,000
Mud fittings on fiberglass insulated pipe	Hall by 1510-15; 12th floor elev. mech. room	940 ea	TSI	Friable 0%	1	O&M; Remove at renovation or demolition	1	No damage	\$70,500
12" vinyl floor tile (all colors) and/or black mastic	1st through 10th floors	38,340 sf	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$268,000
TOTALS								0	\$818,500

Jesse Brown VAMC - Chicago, IL
 Building 11A
 Asbestos Assessments and Cost Estimates
 September 2019

Asbestos Material	Locations	Quantity	AHERA Category	Friability %Damage	AHERA Assessment	Recommendation	Removal Priority	Repair Cost Estimate	Removal Cost Estimate
Pipe insulation	Inaccessible wall and ceiling chases (above/behind hard plaster)	9670 lf	TSI	Friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$531,000
Mud fittings on fiberglass insulated pipe	Inaccessible wall and ceiling chases (above/behind hard plaster)	340 ea	TSI	Friable 0%	1	O&M; Remove at renovation or demolition	1	No damage	\$25,500
12" vinyl floor tile (all colors) and/or black mastic	6th floor	3,640 sf	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$21,840
Laboratory Fume Hoods (Transite)	6219, 6220, 6221, 6256, 6260, 6261, 6273, 6281, 6283, 6285, 7203, 7211, 7218, 7235, 7237, 7239, 7241, 7275, 7281, 7282, 7283, 7287, 7288, 7290, 7288, 7292, 7293, 7298, 7299	2,240 sf	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$56,000
Lab Table Tops, Sinks, and Drying Racks (Ebony Transite)	6219, 6220, 6221, 6256, 6260, 6261, 6273, 6281, 6283, 6285, 7203, 7211, 7218, 7235, 7237, 7239, 7241, 7275, 7281, 7282, 7283, 7287, 7288, 7290, 7288, 7292, 7293, 7298, 7299	5,670 sf	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$56,700
Transite sheets in stairwells	Stairwell 11 to 8th floor penthouse	78 sinks	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$15,600
TOTALS								0	\$706,640

Jesse Brown VAMC - Chicago, IL
 Building 11B
 Asbestos Assessments and Cost Estimates
 September 2019

Asbestos Material	Locations	Quantity	AHERA Category	Friability %Damage	AHERA Assessment	Recommendation	Removal Priority	Repair Cost Estimate	Removal Cost Estimate
Asbestos debris in soil	Crawlspace	2376 sf	TSI	Friable 100%	7	Isolate crawl spaces and place signs at all entrances. Require PPE for entrance to crawl. Encapsulate soil with spray foam or lightweight concrete, or remove soil to depth of 3".	1	\$89,000	\$220,000
12" vinyl floor tile (all colors) and black mastic	3rd floor	3,318 sf	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$25,480
Pipe insulation	Inaccessible wall & ceiling chases	2,000 lf	TSI	Friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$210,000
Mud fittings on fiberglass insulated pipe	Inaccessible wall & ceiling chases	1500 ea	TSI	Friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$30,000
Transite sheets in stairwells	Stairwell 7 (basement and 7th floors)	320 sf	MISC	Non-friable 0%	1	O&M; Remove at renovation or demolition	7	No damage	\$6,000
TOTALS								\$89,000	\$491,480

APPENDIX C

PHOTOGRAPHS OF ASBESTOS MATERIALS

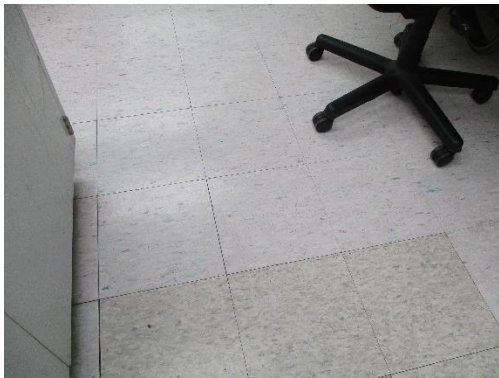
PHOTOGRAPHS OF ASBESTOS MATERIALS BUILDING 1A

Photo 1A-1 Pipe insulation	 A photograph showing three vertical yellow pipes. Each pipe has a white label with the word 'DANGER' in red. To the left of the pipes is a red fire alarm pull station. The background is a light-colored wall.
Photo 1A-2 Pipe insulation	 A photograph showing a complex network of yellow and green pipes. There are several valves and fittings visible. The pipes are running horizontally and vertically. The background is a light-colored wall.
Photo 1A-3 Pipe insulation mud fittings on fiberglass Mud fittings on fiberglass insulated pipe 12" vinyl floor tile (all colors) and/or black mastic	 A photograph showing a green pipe with two green valves. The pipe is lying on a floor that has yellow and black mastic applied to it. The background shows some equipment and a wall.
Photo 1A-4 Pipe insulation and mud fittings on fiberglass	 A photograph showing a yellow staircase. Yellow pipes are running along the wall above the staircase. The background is a light-colored wall.

**PHOTOGRAPHS OF ASBESTOS MATERIALS
BUILDING 1A**

Photo 1A-5 12" vinyl floor tile and/or black mastic	
Photo 1A-6 9" vinyl floor tile and/or black mastic	
Photo 1A-7 12" vinyl floor tile and/or black mastic	
Photo 1A-8 Transite cement sheets behind radiators	

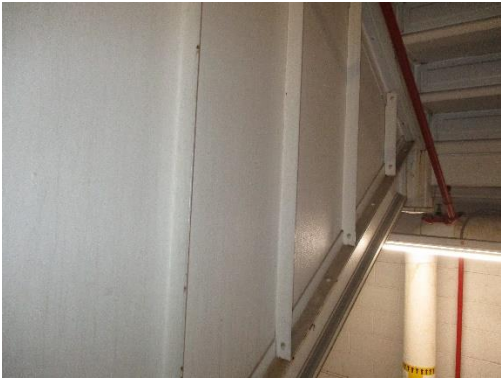
**PHOTOGRAPHS OF ASBESTOS MATERIALS
BUILDING 1B**

Photo 1B-1 Mud fittings on fiberglass insulated pipe	
Photo 1B-2 12" vinyl floor tile and/or black mastic	
Photo 1B-3 12" vinyl floor tile and/or black mastic	
Photo 1B-4 12" vinyl floor tile and/or black mastic	

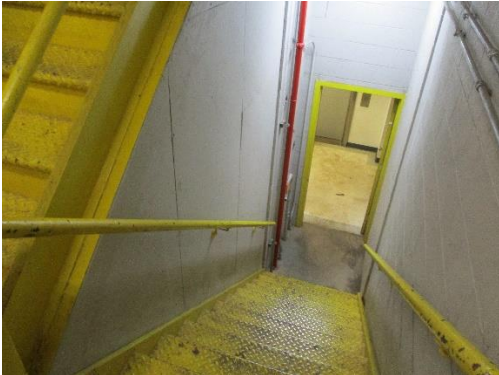
PHOTOGRAPHS OF ASBESTOS MATERIALS BUILDING 11A

Photo 11A-1 12" vinyl floor tile and/or black mastic	
Photo 11A-2 12" vinyl floor tile and/or black mastic	
Photo 11A-3 Laboratory Fume Hoods (Transite)	
Photo 11A-4 Lab Table Tops (Ebony Transite)	

**PHOTOGRAPHS OF ASBESTOS MATERIALS
BUILDING 11A**

Photo 11A-5 Lab Table Sinks and Drying Racks (Ebony Transite)	
Photo 11A-6 Transite sheets in stairwells	
Photo 11A-7 Transite sheets in stairwells	

PHOTOGRAPHS OF ASBESTOS MATERIALS BUILDING 11B

Photo 11B-1 Contaminated soil in crawl space	
Photo 11B-2 Contaminated soil in crawl space	
Photo 11B-3 12" vinyl floor tile and black mastic	
Photo 11B-4 Transite sheets in stairwells	

APPENDIX D

ASBESTOS LABORATORY REPORTS



EMSL Analytical, Inc.

200 Route 130 North Cinnaminson, NJ 08077

Tel/Fax: (800) 220-3675 / (856) 786-5974

<http://www.EMSL.com> / cinnasblab@EMSL.com

EMSL Order: 041926209

Customer ID: CHAS42

Customer PO: JB

Project ID:

Attention: Brett Brumbaugh

Castle Hill Associates, LLC

218-A Mechanic St.

Waterville, OH 43566

Phone: (567) 952-0212

Fax:

Received Date: 09/06/2019 9:10 AM

Analysis Date: 09/14/2019 - 09/16/2019

Collected Date: 09/03/2019 - 09/04/2019

Project: Jesse Brown VAMC / Chicago, IL

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
1 041926209-0001	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	15% Cellulose 10% Glass	15% Vermiculite 60% Non-fibrous (Other)	None Detected
2 041926209-0002	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	20% Cellulose 10% Glass	10% Vermiculite 60% Non-fibrous (Other)	None Detected
3 041926209-0003	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	80% Min. Wool	20% Non-fibrous (Other)	None Detected
4 041926209-0004	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous		85% Non-fibrous (Other)	15% Chrysotile
5 041926209-0005	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	15% Cellulose 15% Glass	70% Non-fibrous (Other)	None Detected
6 041926209-0006	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	15% Cellulose 10% Glass	10% Vermiculite 65% Non-fibrous (Other)	None Detected
7 041926209-0007	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	15% Cellulose 10% Glass	10% Vermiculite 65% Non-fibrous (Other)	None Detected
8 041926209-0008	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	20% Cellulose 15% Glass	10% Vermiculite 55% Non-fibrous (Other)	None Detected
9 041926209-0009	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	20% Cellulose 15% Glass	10% Vermiculite 55% Non-fibrous (Other)	None Detected
10 041926209-0010	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	15% Cellulose 10% Glass	10% Vermiculite 65% Non-fibrous (Other)	None Detected
11 041926209-0011	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	85% Min. Wool	15% Non-fibrous (Other)	None Detected
12 041926209-0012	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	90% Min. Wool	10% Non-fibrous (Other)	None Detected
13 041926209-0013	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	90% Min. Wool	10% Non-fibrous (Other)	None Detected
14 041926209-0014	Bldgs 1A,1B,11A,11B	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
15 041926209-0015	Bldgs 1A,1B,11A,11B	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
16 041926209-0016	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	80% Min. Wool	20% Non-fibrous (Other)	None Detected

Initial report from: 09/14/2019 07:38:45



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EMSL Order: 041926209

Customer ID: CHAS42

Customer PO: JB

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
17	Bldgs 1A, 1B, 11A, 11B	Gray Fibrous Homogeneous	90% Min. Wool	10% Non-fibrous (Other)	None Detected
041926209-0017					
18	Bldgs 1A, 1B, 11A, 11B	Gray Fibrous Homogeneous	90% Min. Wool	10% Non-fibrous (Other)	None Detected
041926209-0018					
19-Floor Tile	Bldgs 1A, 1B, 11A, 11B	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0019					
19-Mastic	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0019A					
20-Floor Tile	Bldgs 1A, 1B, 11A, 11B	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0020					
20-Mastic	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0020A					
21-Floor Tile	Bldgs 1A, 1B, 11A, 11B	Beige Non-Fibrous Homogeneous		97% Non-fibrous (Other)	3% Chrysotile
041926209-0021					
21-Mastic	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		95% Non-fibrous (Other)	5% Chrysotile
041926209-0021A					
22-Floor Tile	Bldgs 1A, 1B, 11A, 11B	Beige Non-Fibrous Homogeneous		97% Non-fibrous (Other)	3% Chrysotile
041926209-0022					
22-Mastic	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		95% Non-fibrous (Other)	5% Chrysotile
041926209-0022A					
23-Floor Tile	Bldgs 1A, 1B, 11A, 11B	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0023					
23-Mastic	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0023A					
24-Floor Tile	Bldgs 1A, 1B, 11A, 11B	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0024					
24-Mastic	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0024A					
25-Floor Tile	Bldgs 1A, 1B, 11A, 11B	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0025					
25-Mastic	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0025A					
26-Floor Tile	Bldgs 1A, 1B, 11A, 11B	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0026					
26-Mastic	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0026A					
27-Floor Tile	Bldgs 1A, 1B, 11A, 11B	Green Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0027					

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<http://www.EMSL.com> / cinnasblab@EMSL.com

EMSL Order: 041926209

Customer ID: CHAS42

Customer PO: JB

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
27-Mastic	Bldgs 1A,1B,11A,11B	Gray/Black Non-Fibrous Heterogeneous		100% Non-fibrous (Other)	None Detected
041926209-0027A Mastic and leveler are inseparable, sample analyzed as a composite					
28-Floor Tile	Bldgs 1A,1B,11A,11B	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0028					
28-Mastic	Bldgs 1A,1B,11A,11B	Yellow Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0028A					
29	Bldgs 1A,1B,11A,11B	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0029					
30	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	90% Min. Wool	10% Non-fibrous (Other)	None Detected
041926209-0030					
31	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	85% Min. Wool	15% Non-fibrous (Other)	None Detected
041926209-0031					
32	Bldgs 1A,1B,11A,11B	Gray Fibrous Homogeneous	90% Min. Wool	10% Non-fibrous (Other)	None Detected
041926209-0032					
40-Floor Tile	Bldgs 1A,1B,11A,11B	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0033					
40-Mastic	Bldgs 1A,1B,11A,11B	Gray/Black Non-Fibrous Heterogeneous		96% Non-fibrous (Other)	4% Chrysotile
041926209-0033A Mastic and leveler are inseparable, sample analyzed as a composite					
41-Floor Tile	Bldgs 1A,1B,11A,11B	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0034					
41-Mastic	Bldgs 1A,1B,11A,11B	Gray/Black Non-Fibrous Heterogeneous		97% Non-fibrous (Other)	3% Chrysotile
041926209-0034A Mastic and leveler are inseparable, sample analyzed as a composite					
42-Floor Tile	Bldgs 1A,1B,11A,11B	Beige Non-Fibrous Homogeneous		98% Non-fibrous (Other)	2% Chrysotile
041926209-0035					
42-Mastic	Bldgs 1A,1B,11A,11B	Black Non-Fibrous Homogeneous		95% Non-fibrous (Other)	5% Chrysotile
041926209-0035A					
43-Floor Tile	Bldgs 1A,1B,11A,11B	Beige Non-Fibrous Homogeneous		97% Non-fibrous (Other)	3% Chrysotile
041926209-0036					
43-Mastic	Bldgs 1A,1B,11A,11B	Black Non-Fibrous Homogeneous		96% Non-fibrous (Other)	4% Chrysotile
041926209-0036A					
44-Floor Tile	Bldgs 1A,1B,11A,11B	Green Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0037					
44-Mastic	Bldgs 1A,1B,11A,11B	Black Non-Fibrous Homogeneous	3% Cellulose	95% Non-fibrous (Other)	2% Chrysotile
041926209-0037A					
45-Floor Tile	Bldgs 1A,1B,11A,11B	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
041926209-0038					

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Tel/Fax: (800) 220-3675 / (856) 786-5974

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EMSL Order: 041926209

Customer ID: CHAS42

Customer PO: JB

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
45-Mastic 041926209-0038A	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
46-Floor Tile 041926209-0039	Bldgs 1A, 1B, 11A, 11B	Beige Non-Fibrous Homogeneous		97% Non-fibrous (Other)	3% Chrysotile
46-Mastic 041926209-0039A	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		96% Non-fibrous (Other)	4% Chrysotile
47-Floor Tile 041926209-0040	Bldgs 1A, 1B, 11A, 11B	White/Black Non-Fibrous Homogeneous		93% Non-fibrous (Other)	7% Chrysotile
47-Mastic 041926209-0040A	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		95% Non-fibrous (Other)	5% Chrysotile
48-Floor Tile 041926209-0041	Bldgs 1A, 1B, 11A, 11B	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
48-Mastic 041926209-0041A	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		97% Non-fibrous (Other)	3% Chrysotile
49-Floor Tile 041926209-0042	Bldgs 1A, 1B, 11A, 11B	Gray/White Non-Fibrous Homogeneous		96% Non-fibrous (Other)	4% Chrysotile
49-Mastic 041926209-0042A	Bldgs 1A, 1B, 11A, 11B	Black/Yellow Non-Fibrous Heterogeneous	4% Cellulose	93% Non-fibrous (Other)	3% Chrysotile
50-Floor Tile 041926209-0043	Bldgs 1A, 1B, 11A, 11B	Tan/White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
50-Mastic 041926209-0043A	Bldgs 1A, 1B, 11A, 11B	Black/Yellow Non-Fibrous Heterogeneous		97% Non-fibrous (Other)	3% Chrysotile
51-Floor Tile 041926209-0044	Bldgs 1A, 1B, 11A, 11B	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
51-Mastic 041926209-0044A	Bldgs 1A, 1B, 11A, 11B	Yellow Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
52-Floor Tile 041926209-0045	Bldgs 1A, 1B, 11A, 11B	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
52-Mastic 041926209-0045A	Bldgs 1A, 1B, 11A, 11B	Black/Yellow Non-Fibrous Heterogeneous		97% Non-fibrous (Other)	3% Chrysotile
53-Floor Tile 041926209-0046	Bldgs 1A, 1B, 11A, 11B	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
53-Mastic 041926209-0046A	Bldgs 1A, 1B, 11A, 11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
54-Floor Tile 041926209-0047	Bldgs 1A, 1B, 11A, 11B	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
54-Mastic 041926209-0047A	Bldgs 1A, 1B, 11A, 11B	White/Black/Yellow Non-Fibrous Heterogeneous	3% Cellulose	97% Non-fibrous (Other)	None Detected

Initial report from: 09/14/2019 07:38:45



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<http://www.EMSL.com> / cinnasblab@EMSL.com

EMSL Order: 041926209

Customer ID: CHAS42

Customer PO: JB

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
Mastic and leveler are inseparable, sample analyzed as a composite.					
55-Floor Tile 041926209-0048	Bldgs 1A,1B,11A,11B	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
55-Mastic 041926209-0048A	Bldgs 1A,1B,11A,11B	Yellow Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
55-Floor Tile 2 041926209-0048B	Bldgs 1A,1B,11A,11B	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
55-Mastic 2 041926209-0048C	Bldgs 1A,1B,11A,11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
56-Floor Tile 041926209-0049	Bldgs 1A,1B,11A,11B	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
56-Mastic 041926209-0049A	Bldgs 1A,1B,11A,11B	Black Non-Fibrous Homogeneous		98% Non-fibrous (Other)	2% Chrysotile
57-Floor Tile 041926209-0050	Bldgs 1A,1B,11A,11B	Tan/Beige Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
57-Mastic 041926209-0050A	Bldgs 1A,1B,11A,11B	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected

Analyst(s)

Amy Johnson (50)

Olufunke Akintunde (24)

Seri Smith (6)

Benjamin Ellis, Laboratory Manager
or Other Approved Signatory

EMSL maintains liability limited to cost of analysis. The above analyses were performed in general compliance with Appendix E to Subpart E of 40 CFR (previously EPA 600/M4-82-020 "Interim Method"), but augmented with procedures outlined in the 1993 ("final") version of the method. This report relates only to the samples reported above, and may not be reproduced, except in full, without written approval by EMSL. EMSL bears no responsibility for sample collection activities or analytical method limitations. Interpretation and use of test results are the responsibility of the client. All samples received in acceptable condition unless otherwise noted. This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST or any agency of the federal government. EMSL recommends gravimetric reduction for all non-friable organically bound materials prior to analysis. Estimation of uncertainty is available on request.

Samples analyzed by EMSL Analytical, Inc. Cinnaminson, NJ NVLAP Lab Code 101048-0, AIHA-LAP, LLC-IHLAP Lab 100194, NYS ELAP 10872, NJ DEP 03036, PA ID# 68-00367, LA #04127

Initial report from: 09/14/2019 07:38:45

**EMSL Analytical, Inc.**

200 Route 130 North, Cinnaminson, NJ 08077

Phone/Fax: (800) 220-3675 / (856) 786-5974

<http://www.EMSL.com>cinnaslab@EMSL.com

EMSL Order: 041926203

CustomerID: CHAS42

CustomerPO: JB

ProjectID:

Attn: **Brett Brumbaugh**
Castle Hill Associates, LLC
218-A Mechanic St.
Waterville, OH 43566

Phone: (567) 952-2012
Fax:
Received: 09/06/19 9:10 AM
Analysis Date: 9/11/2019
Collected: 9/4/2019

Project: **Jess Brown VAMC / Chicago, IL****Test Report: Qualitative asbestos analysis of soils using the EPA 600/R-93/116 method**

Sample	Description	Appearance	Result	Notes
33 041926203-0001	Building 11A - Crawl Space Soil	Brown Fibrous Homogeneous	None Detected	
34 041926203-0002	Building 11A - Crawl Space Soil	Brown Fibrous Homogeneous	None Detected	
35 041926203-0003	Building 11A - Crawl Space Soil	Brown/White Fibrous Homogeneous	None Detected	
36 041926203-0004	Building 11B - Crawl Space Soil	Brown Fibrous Homogeneous	Chrysotile	
37 041926203-0005	Building 11B - Crawl Space Soil	Brown Fibrous Homogeneous	None Detected	
38 041926203-0006	Building 11B - Crawl Space Soil	Brown Fibrous Homogeneous	Chrysotile	
39 041926203-0007	Building 11B - Crawl Space Soil	Brown Fibrous Homogeneous	None Detected	
58 041926203-0008	Building 11B - Crawl Space Soil	Brown Fibrous Homogeneous	Chrysotile	

Qualitative analysis of soil samples without matrix modification does not include the examination of the entire sample material for the identification of asbestos. Data from these samples may be biased low and subject to false negative results.

Analyst(s)

Will DiBella (8)

Benjamin Ellis, Laboratory Manager
or other approved signatory

Soil is a problem matrix due to its inherent heterogeneity and there is a likelihood for false negatives with this analysis. EMSL recommends more specialized methodologies such as the EPA 600/R-93/116 with milling preparation. EMSL maintains liability limited to cost of analysis. This report relates only to the samples reported above and may not be reproduced, except in full, without written approval by EMSL. EMSL bears no responsibility for sample collection activities or analytical method limitations. Interpretation and use of test results are the responsibility of the client. Samples received in good condition unless otherwise noted.

Samples analyzed by EMSL Analytical, Inc. Cinnaminson, NJ

Initial report from 09/11/2019 22:35:11



EMSL ANALYTICAL, INC.
LABORATORY • PRODUCTS • TRAINING

Asbestos Chain of Custody

EMSL Order Number (Lab Use Only):

041926209

EMSL Analytical, Inc.
212 South Wagner Road

RECEIVED
Andover, MA 01810
PHONE: (734) 668-6810
FAX: (734) 668-8532
2019 SEP 5 AM 9:11

Company Name : Castle Hill Associates, LLC		EMSL Customer ID:	
Street: 218-A Mechanic St.		City: Waterville	State/Province: OH
Zip/Postal Code: 43566	Country: US	Telephone #: 567-952-2012	Fax #:
Report To (Name): Brett Brumbaugh		Please Provide Results: ☐ Fax ☒ Email	
Email Address: bbrumbaugh@castlehillassociates.com		Purchase Order: JB	
Project Name/Number: Jesse Brown VAMC - Chicago		EMSL Project ID (Internal Use Only):	
U.S. State Samples Taken: IL		CT Samples: ☐ Commercial/Taxable ☐ Residential/Tax Exempt	

EMSL-Bill to: ☒ Same ☐ Different - If Bill to is Different note instructions in Comments**

Third Party Billing requires written authorization from third party

Turnaround Time (TAT) Options* - Please Check

☐ 3 Hour ☐ 6 Hour ☐ 24 Hour ☐ 48 Hour ☐ 72 Hour ☐ 96 Hour ☒ 1 Week ☐ 2 Week

*For TEM Air 3 hr through 6 hr, please call ahead to schedule. *There is a premium charge for 3 Hour TEM AHERA or EPA Level II TAT. You will be asked to sign an authorization form for this service. Analysis completed in accordance with EMSL's Terms and Conditions located in the Analytical Price Guide.

PCM - Air ☐ Check if samples are from NY

☐ NIOSH 7400
☐ w/ OSHA 8hr. TWA

PLM - Bulk (reporting limit)

☒ PLM EPA 600/R-93/116 (<1%)

☐ PLM EPA NOB (<1%)

Point Count

☐ 400 (<0.25%) ☐ 1000 (<0.1%)

Point Count w/Gravimetric

☐ 400 (<0.25%) ☐ 1000 (<0.1%) -

☐ NYS 198.1 (friable in NY)

☐ NYS 198.6 NOB (non-friable-NY)

☐ NYS 198.8 SOF-V

☐ NIOSH 9002 (<1%)

TEM - Air ☐ 4-4.5hr TAT (AHERA only)

☐ AHERA 40 CFR, Part 763

☐ NIOSH 7402

☐ EPA Level II

☐ ISO 10312

TEM - Bulk

☐ TEM EPA NOB

☐ NYS NOB 198.4 (non-friable-NY)

☐ Chatfield SOP

☐ TEM Mass Analysis-EPA 600 sec. 2.5

TEM - Water: EPA 100.2

Fibers >10µm ☐ Waste ☐ Drinking

All Fiber Sizes ☐ Waste ☐ Drinking

TEM- Dust

☐ Microvac - ASTM D 5755

☐ Wipe - ASTM D6480

☐ Carpet Sonication (EPA 600/J-93/167)

Soil/Rock/Vermiculite

☐ PLM EPA 600/R-93/116 with milling prep (<1%)

☐ PLM EPA 600/R-93/116 with milling prep (<0.25%)

☐ TEM EPA 600/R-93/116 with milling prep (<0.1%)

☐ TEM Qualitative via Filtration Prep

☐ TEM Qualitative via Drop Mount Prep

☐ Cincinnati Method EPA 600/R-04/004 - PLM/TEM (BC only)

Other:

☐

☐ Check For Positive Stop - Clearly Identify Homogenous Group

Filter Pore Size (Air Samples): ☐ 0.8µm ☐ 0.45µm

Samplers Name: BRETT BRUMBAUGH

Samplers Signature:

Sample #	Sample Description	Volume/Area (Air) HA # (Bulk)	Date/Time Sampled
1-32	Bldgs. 1A, 1B, 11A, 11B		9/3/19
40-57	" " " " "		9/4/19

Client Sample # (s): 1-32, 40-57

Total # of Samples: 50

Relinquished (Client):

Date: 9-5-19

Time: 12:35

Received (Lab):

Date: 9-6-19

Time: 9:10

Comments/Special Instructions:

50



EMSL ANALYTICAL, INC.
LABORATORY • PRODUCTS • TRAINING

Asbestos Chain of Custody

EMSL Order Number (Lab Use Only):

041926203

EMSL Analytical, Inc.
212 South Wagner Road

RECEIVED
Cincinnati, OH 45210
PHONE: (784) 668-6810
FAX: (734) 668-8532
2019 SEP -6 AM 9:11

Company Name: Castle Hill Associates, LLC		EMSL Customer ID:	
Street: 218-A Mechanic St.		City: Waterville	State/Province: OH
Zip/Postal Code: 43566	Country: US	Telephone #: 567-952-2012	Fax #:
Report To (Name): Brett Brumbaugh		Please Provide Results: ☐ Fax ☒ Email	
Email Address: bbrumbaugh@castlehillassociates.com		Purchase Order: JB	
Project Name/Number: Jesse Brown VAMC - Chicago		EMSL Project ID (Internal Use Only):	
U.S. State Samples Taken: IL		CT Samples: ☐ Commercial/Taxable ☐ Residential/Tax Exempt	
EMSL-Bill to: ☒ Same ☐ Different - If Bill to is Different note instructions in Comments** Third Party Billing requires written authorization from third party			
Turnaround Time (TAT) Options* - Please Check			
☐ 3 Hour ☐ 6 Hour ☐ 24 Hour ☐ 48 Hour ☐ 72 Hour ☐ 96 Hour ☒ 1 Week ☐ 2 Week			
*For TEM Air 3 hr through 6 hr, please call ahead to schedule. *There is a premium charge for 3 Hour TEM AHERA or EPA Level II TAT. You will be asked to sign an authorization form for this service. Analysis completed in accordance with EMSL's Terms and Conditions located in the Analytical Price Guide.			
PCM - Air ☐ Check if samples are from NY ☐ NIOSH 7400 ☐ w/ OSHA 8hr. TWA		TEM - Air ☐ 4-4.5hr TAT (AHERA only) ☐ AHERA 40 CFR, Part 763 ☐ NIOSH 7402 ☐ EPA Level II ☐ ISO 10312	
PLM - Bulk (reporting limit) ☐ PLM EPA 600/R-93/116 (<1%) ☐ PLM EPA NOB (<1%) Point Count ☐ 400 (<0.25%) ☐ 1000 (<0.1%) Point Count w/Gravimetric ☐ 400 (<0.25%) ☐ 1000 (<0.1%) ☐ NYS 198.1 (friable in NY) ☐ NYS 198.6 NOB (non-friable-NY) ☐ NYS 198.8 SOF-V ☐ NIOSH 9002 (<1%)		TEM - Bulk ☐ TEM EPA NOB ☐ NYS NOB 198.4 (non-friable-NY) ☐ Chatfield SOP ☐ TEM Mass Analysis-EPA 600 sec. 2.5 TEM - Water: EPA 100.2 Fibers >10µm ☐ Waste ☐ Drinking All Fiber Sizes ☐ Waste ☐ Drinking	
TEM - Dust ☐ Microvac - ASTM D 5755 ☐ Wipe - ASTM D6480 ☐ Carpet Sonication (EPA 600/J-93/167)		Soil/Rock/Vermiculite ☐ PLM EPA 600/R-93/116 with milling prep (<1%) ☐ PLM EPA 600/R-93/116 with milling prep (<0.25%) ☐ TEM EPA 600/R-93/116 with milling prep (<0.1%) ☐ TEM Qualitative via Filtration Prep ☐ TEM Qualitative via Drop Mount Prep ☐ Cincinnati Method EPA 600/R-04/004 - PLM/TEM (BC only)	
☐ Check For Positive Stop - Clearly Identify Homogenous Group		Filter Pore Size (Air Samples): ☐ 0.8µm ☐ 0.45µm	
Samplers Name: BRETT BRUMBAUGH		Samplers Signature:	
Sample #	Sample Description	Volume/Area (Air) HA # (Bulk)	Date/Time Sampled
33-35	Building 11A crawl space soil		9/4/19
36-39	Building 11B crawl space soil		9/4/19
58	Building 11B crawl space soil		9/4/19
Client Sample # (s): 33-39 and 58 - Total # of Samples: 8			
Relinquished (Client):		Date: 9-5-19	Time: 3:35
Received (Lab):		Date: 9-6-19	Time: 9:10
Comments/Special Instructions:			

8.

APPENDIX E

INSPECTOR CERTIFICATIONS



525-535 West Jefferson Street • Springfield, Illinois 62761-0001 • www.dph.illinois.gov

BRETT L BRUMBAUGH
7920 N WOODBRIDGE ROAD
MONCLOVA, OH 43542

3/15/2019

ASBESTOS PROFESSIONAL LICENSE ID NUMBER: 09972

Enclosed is your Asbestos Professional License. Please note the expiration date on the card and in the image depicted below.

COPY OF THE ASBESTOS PROFESSIONAL LICENSE

Front of License

Back of License

		ASBESTOS PROFESSIONAL LICENSE		ENDORSEMENTS	TC EXPIRES
				INSPECTOR	10/24/2019
ID NUMBER	ISSUED	EXPIRES			
100 - 09972	3/15/2019	05/15/2020			
BRETT L BRUMBAUGH 7920 N WOODBRIDGE ROAD MONCLOVA, OH 43542 Environmental Health				Alteration of this license shall result in legal action This license issued under authority of the State of Illinois Department of Public Health This license is valid only when accompanied by a valid training course certificate.	

If you have any questions or need further assistance, contact the Asbestos Program at (217)782-3517 or fax (217)785-5897.

Our WEB address is: dph.illinois.gov/topics-services/environmental-health-protection/asbestos
EMAIL Address: dph.asbestos@illinois.gov

APPENDIX F

LAB CERTIFICATIONS

United States Department of Commerce
National Institute of Standards and Technology



Certificate of Accreditation to ISO/IEC 17025:2017

NVLAP LAB CODE: 101048-0

EMSL Analytical, Inc.
Cinnaminson, NJ

*is accredited by the National Voluntary Laboratory Accreditation Program for specific services,
listed on the Scope of Accreditation, for:*

Asbestos Fiber Analysis

*This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2017.
This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality
management system (refer to joint ISO-ILAC-IAF Communique dated January 2009).*

2019-07-01 through 2020-06-30

Effective Dates



Dana S. Laman
For the National Voluntary Laboratory Accreditation Program

SCOPE OF ACCREDITATION TO ISO/IEC 17025:2017

EMSL Analytical, Inc.
200 Route 130 North
Cinnaminson, NJ 08077
Mr. Ben Ellis
Phone: 800-220-3675 Fax: 856-786-5973
Email: bellis@emsl.com
<http://www.emsl.com>

ASBESTOS FIBER ANALYSIS

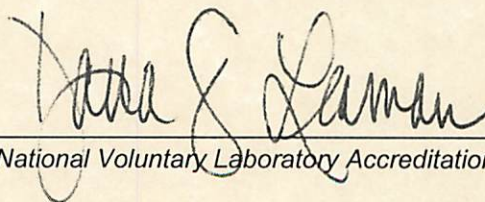
NVLAP LAB CODE 101048-0

Bulk Asbestos Analysis

<u>Code</u>	<u>Description</u>
18/A01	EPA -- 40 CFR Appendix E to Subpart E of Part 763, Interim Method of the Determination of Asbestos in Bulk Insulation Samples
18/A03	EPA 600/R-93/116: Method for the Determination of Asbestos in Bulk Building Materials

Airborne Asbestos Analysis

<u>Code</u>	<u>Description</u>
18/A02	U.S. EPA's "Interim Transmission Electron Microscopy Analytical Methods-Mandatory and Nonmandatory-and Mandatory Section to Determine Completion of Response Actions" as found in 40 CFR, Part 763, Subpart E, Appendix A.



For the National Voluntary Laboratory Accreditation Program

APPENDIX G

ASBESTOS LOCATIONS & SAMPLE LOCATION DRAWINGS

DEPARTMENT OF VETERANS AFFAIRS

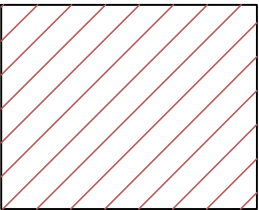
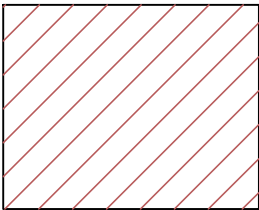
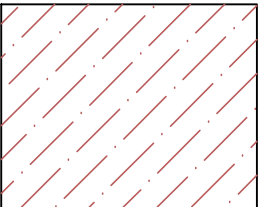
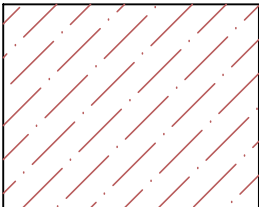
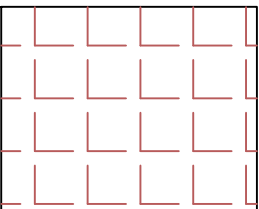
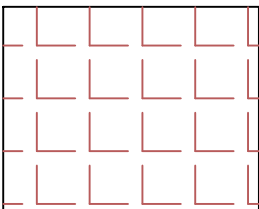
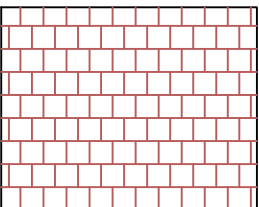
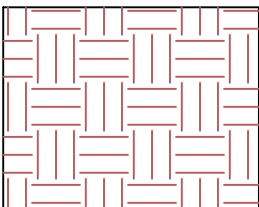
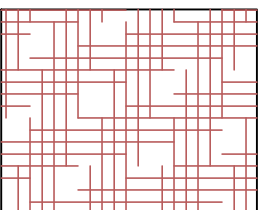
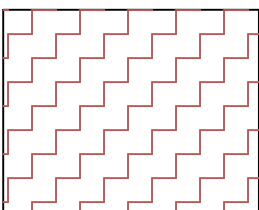
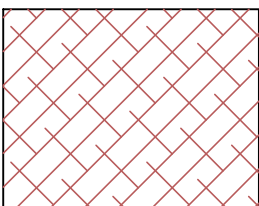
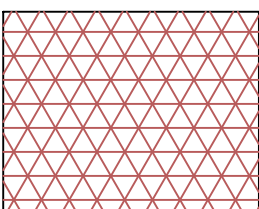
JESSE BROWN VA MEDICAL CENTER

CHICAGO, ILLINOIS

2019 ASBESTOS REINSPECTION

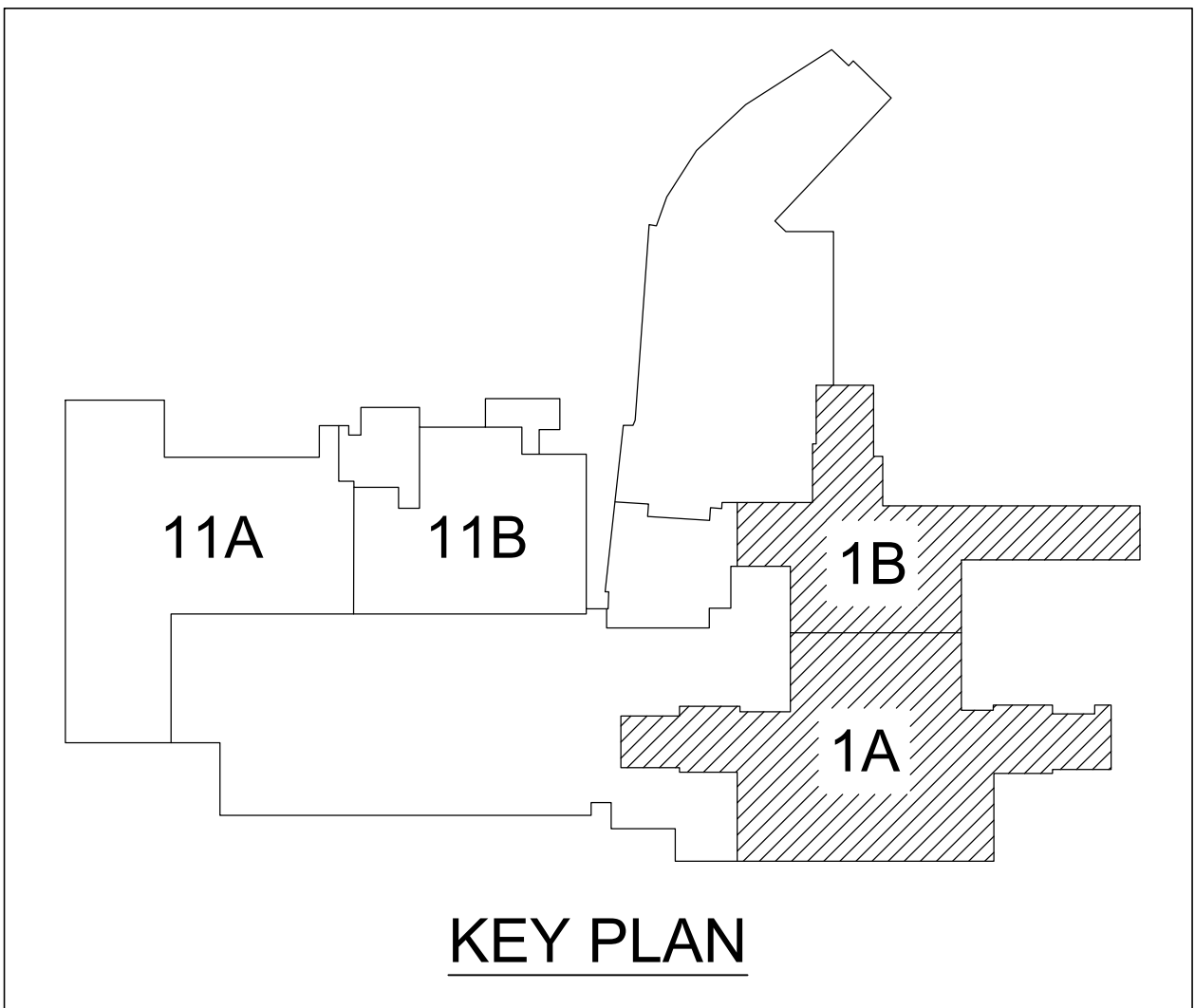
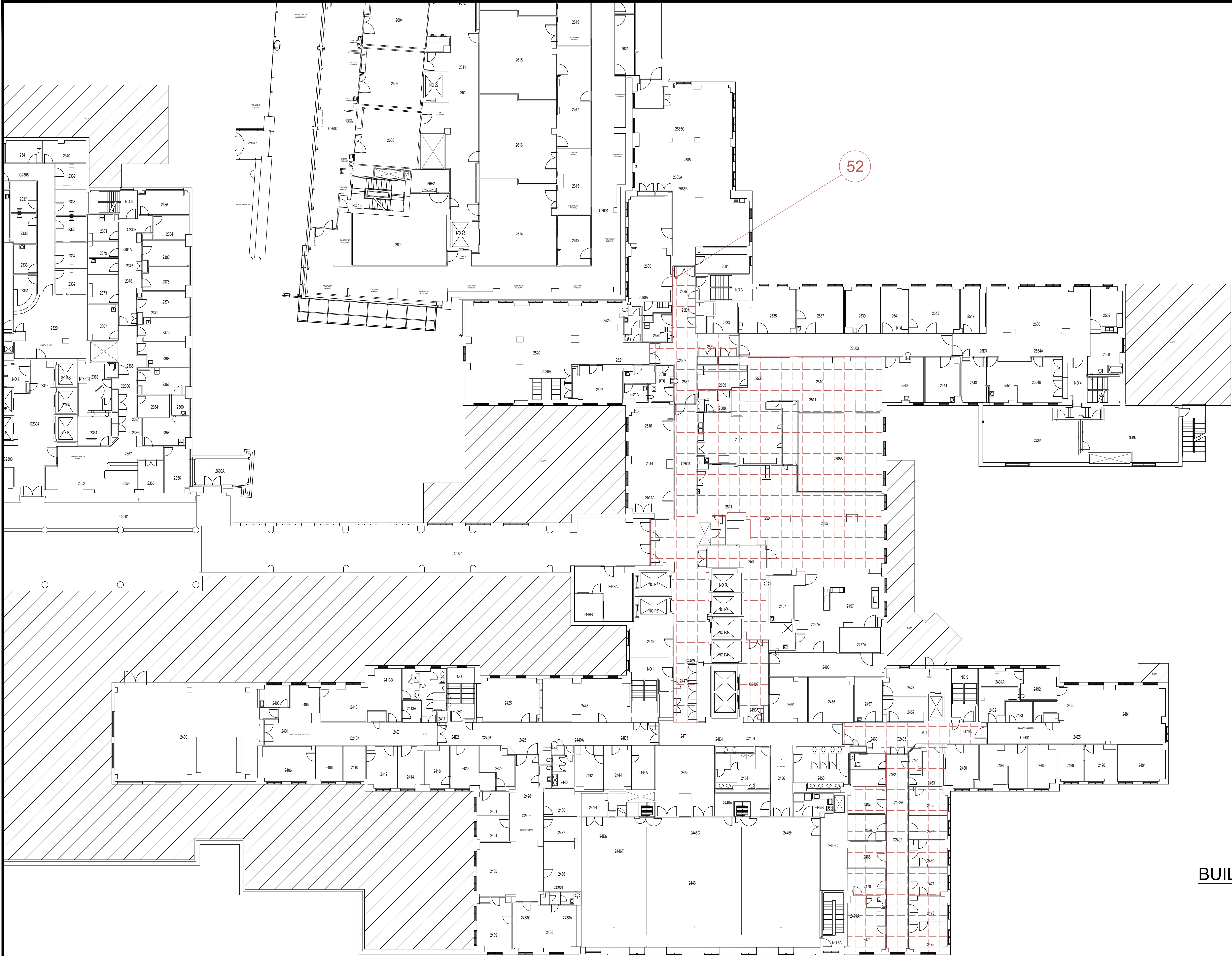
VA CONTRACT NUMBER: 36C24619N0872
TASK ORDER: 36C25219F0327

LEGEND

BUILDING 1 A&B		BUILDING 11 A&B	
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	ASBESTOS-CONTAINING PIPE INSULATION		ASBESTOS-CONTAINING PIPE INSULATION
	ASBESTOS-CONTAINING MUDDIED FITTINGS ON FIBERGLASS INSULATED PIPES		ASBESTOS-CONTAINING MUDDIED FITTINGS ON FIBERGLASS INSULATED PIPES
	ASBESTOS-CONTAINING 12" VINYL FLOOR TILE (ALL COLORS) AND/OR BLACK MASTIC		ASBESTOS-CONTAINING 12" VINYL FLOOR TILE (ALL COLORS) AND/OR BLACK MASTIC
	ASBESTOS-CONTAINING 9" VINYL FLOOR TILE AND BLACK MASTIC		ASBESTOS-CONTAINING PIPE INSULATION DEBRIS IN SOIL/DIRT IN CRAWL SPACE
	ASBESTOS-CONTAINING TRANSITE CEMENT SHEETS BEHIND RADIATORS		ASBESTOS-CONTAINING TRANSITE LABORATORY FUME HOODS
			ASBESTOS-CONTAINING EBONY TRANSITE LAB TABLE TOPS, SINKS, AND DRYING RACKS
			ASBESTOS-CONTAINING TRANSITE SHEETS IN STAIRWELLS

Revisions	Dates	ENVIRONMENTAL CONSULTANT: CASTLE HILL ASSOCIATES, LLC. 218-A MECHANIC STREET, WATERVILLE, OHIO 43566 Tel: (567) 952-0212		Professional Seal: 	North:	Drawing Title TITLE SHEET	Project Title FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS	Date SEPTEMBER 2019		Department of Veterans Affairs 			
						Approved : Division Chief	Building Number -	Checked BLB			Drawn AFS	Project No. 537-19-122	Drawing No. CS
						Approved : Service Director	Location CHICAGO, ILLINOIS				Sheet 01 of 24		

three inches = one foot
one and one-half inches = one foot
one inch = one foot
three-quarters inch = one foot
one-half inch = one foot
three-eighths inch = one foot
one-quarter inch = one foot
one-eighth inch = one foot



BUILDING # 1A & 1B - SECOND FLOOR PLAN

LEGEND

ASBESTOS-CONTAINING 12" VINYL FLOOR TILE (ALL COLORS) AND/OR BLACK MASTIC

Revisions	Dates

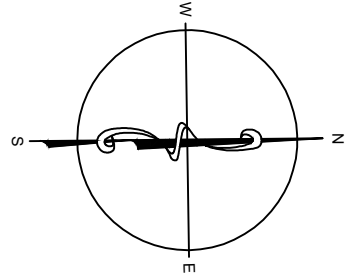
ENVIRONMENTAL CONSULTANT:
CASTLE HILL ASSOCIATES, LLC.
219-A MECHANIC STREET,
WATERVILLE, OHIO 43566
Tel: (567) 952-0212



Professional Seal:



North:



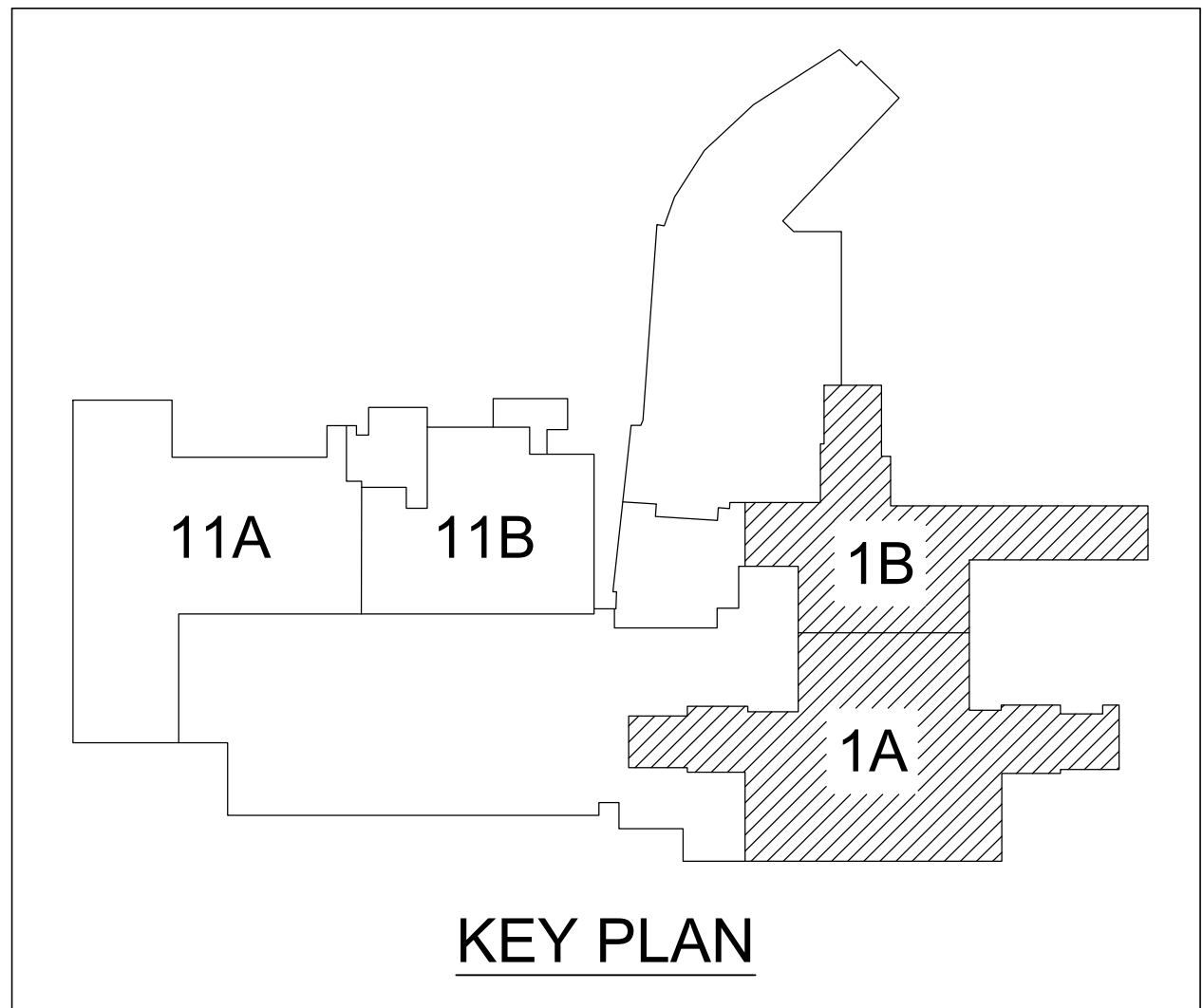
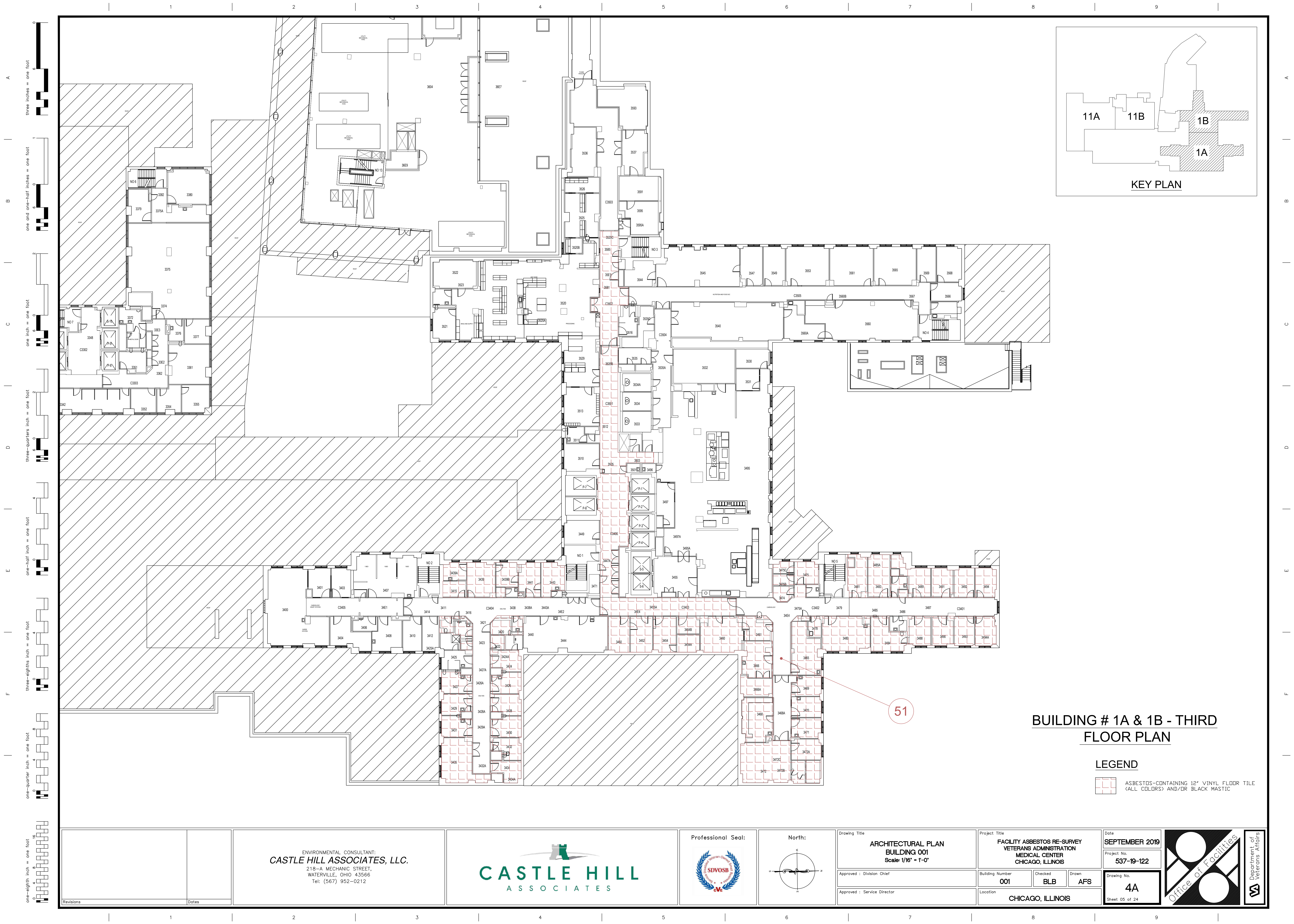
Drawing Title
ARCHITECTURAL PLAN BUILDING 001 Scale: 1/16" = 1'-0"
Approved : Division Chief
Approved : Service Director

Project Title
FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS
Building Number
001
Location
CHICAGO, ILLINOIS

Date
SEPTEMBER 2019
Project No.
537-19-122
Drawing No.
3A
Sheet 04 of 24



Department of
Veterans Affairs

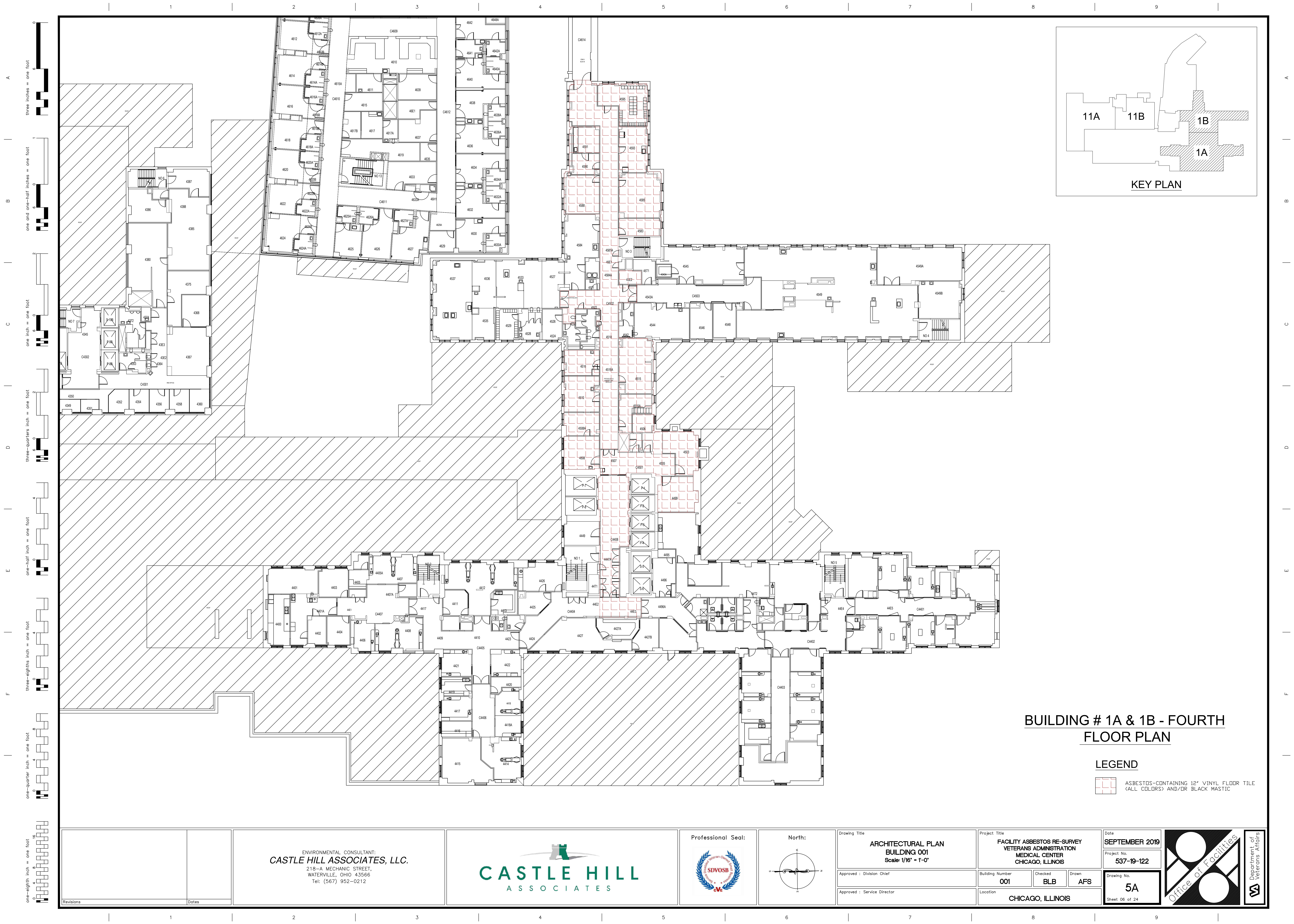


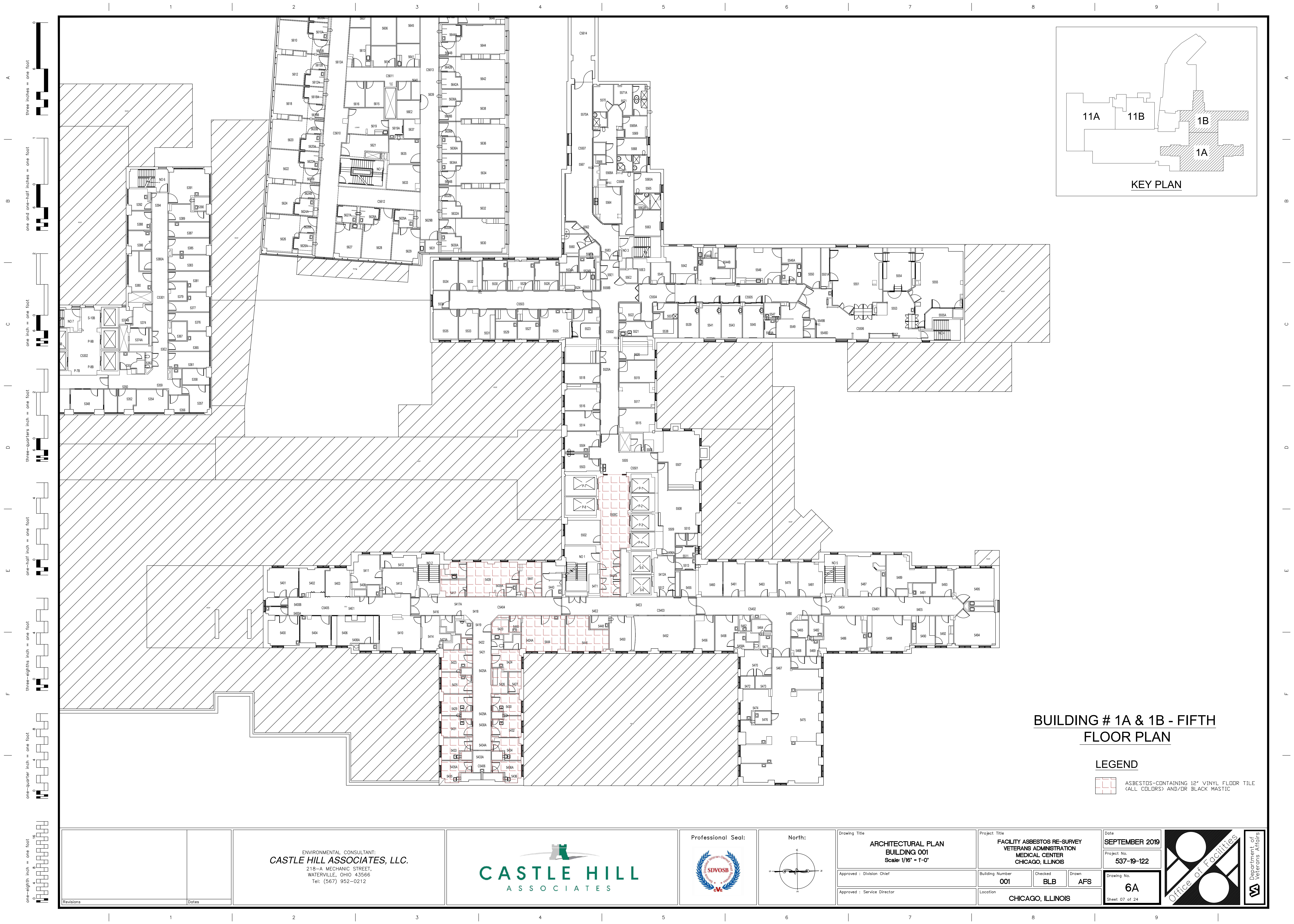
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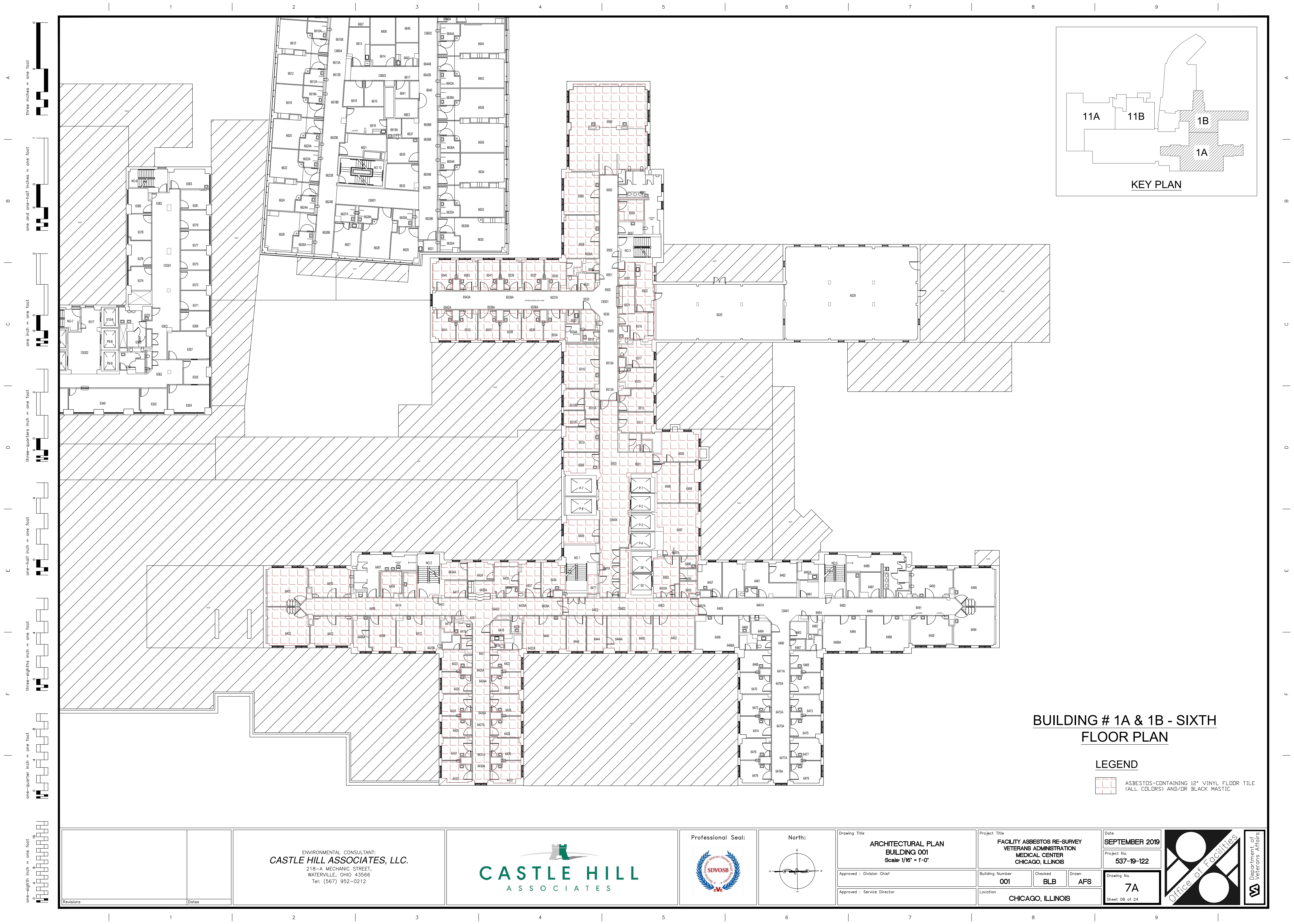
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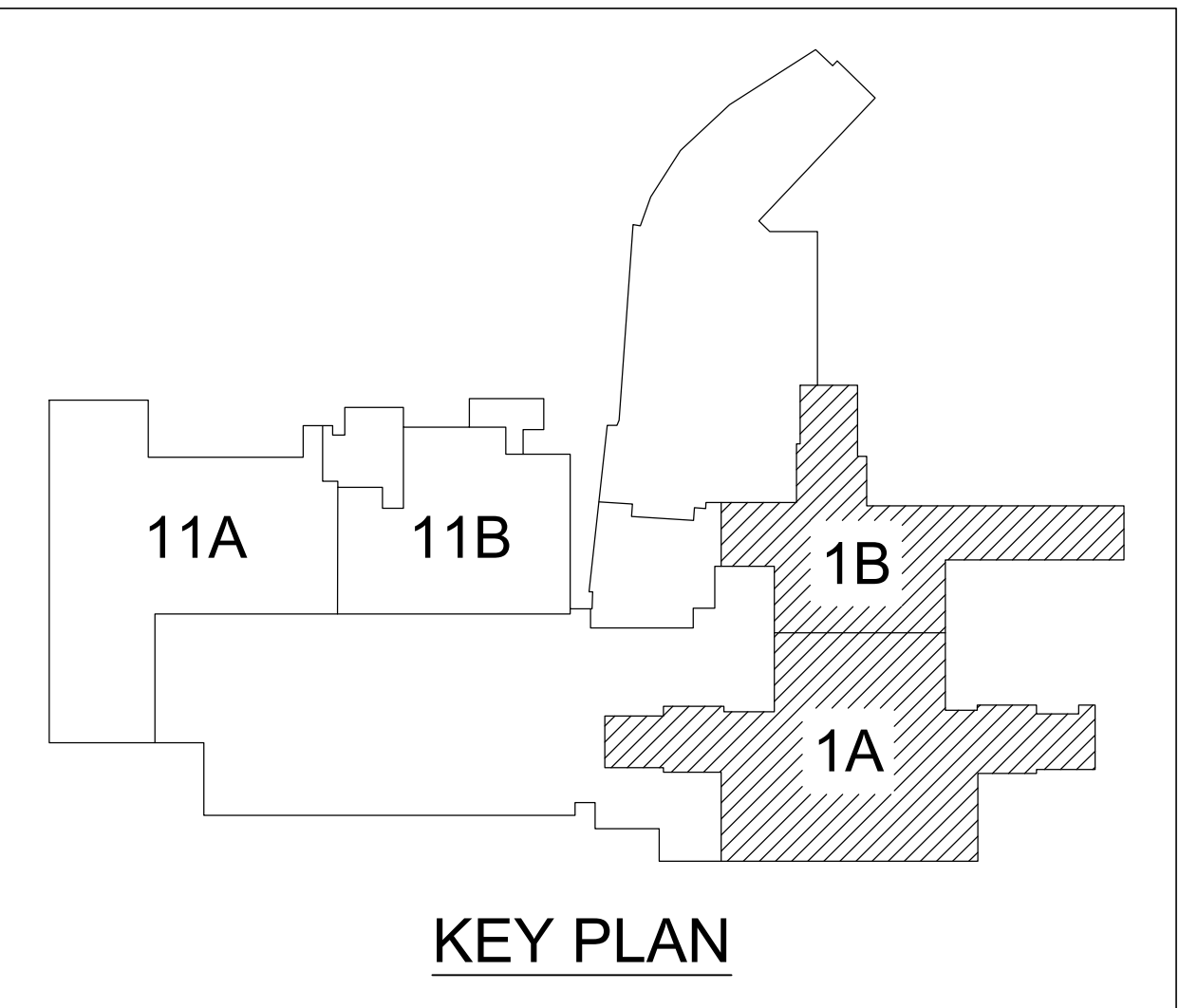
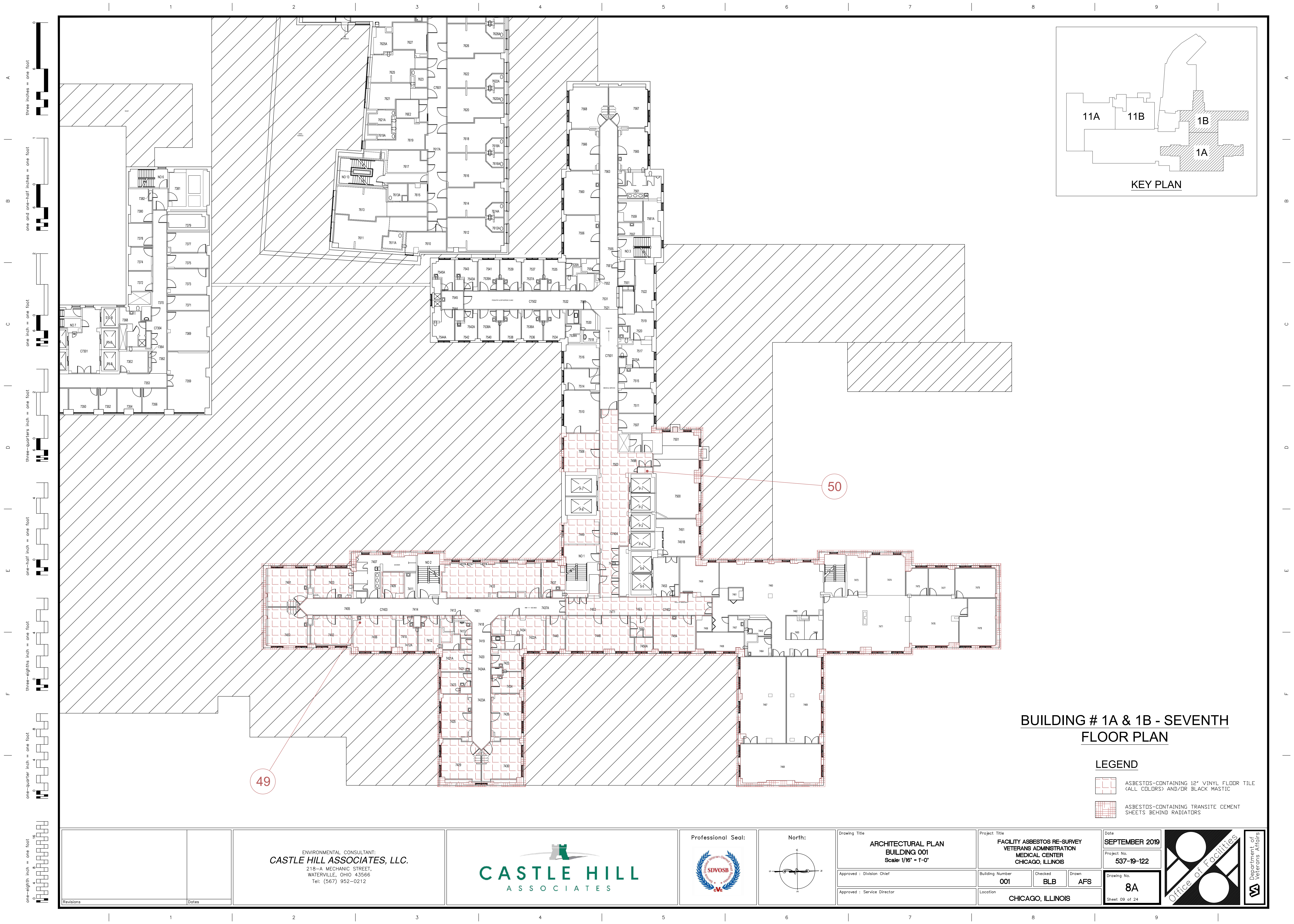
ASBESTOS-CONTAINING 12" VINYL FLOOR TILE (ALL COLORS) AND/OR BLACK MASTIC

Revisions	Dates	ENVIRONMENTAL CONSULTANT: CASTLE HILL ASSOCIATES, LLC. 218-A MECHANIC STREET, WATERVILLE, OHIO 43566 Tel: (567) 952-0212		Professional Seal: 	North: 	Drawing Title ARCHITECTURAL PLAN BUILDING 001 Scale: 1/16" = 1'-0"	Project Title FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS	Date SEPTEMBER 2019		Department of Veterans Affairs		
						Approved: Division Chief	Building Number 001	Checked BLB			Drawn AFS	Project No. 537-19-122
						Approved: Service Director	Location CHICAGO, ILLINOIS	Drawing No. 4A			Sheet 05 of 24	







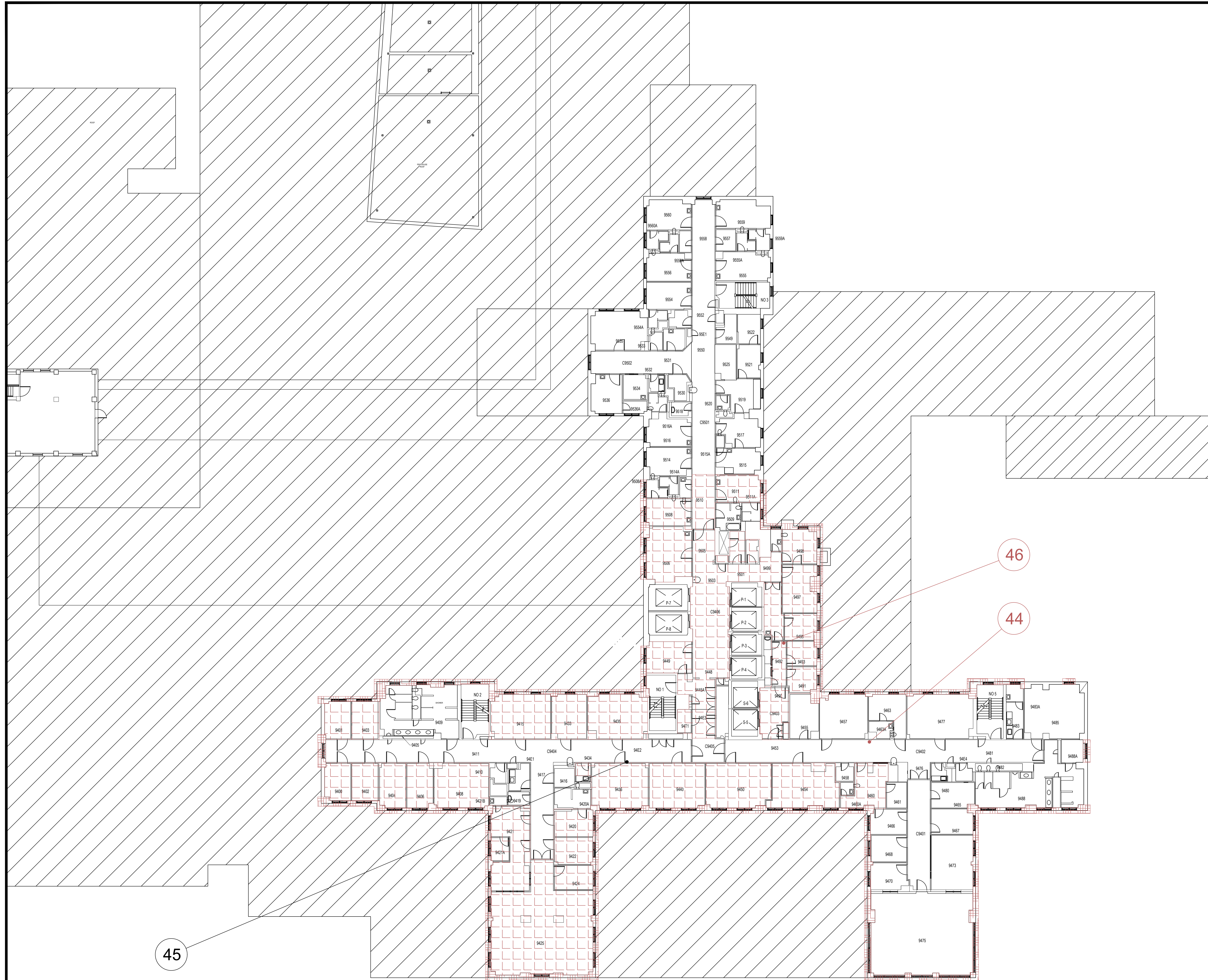
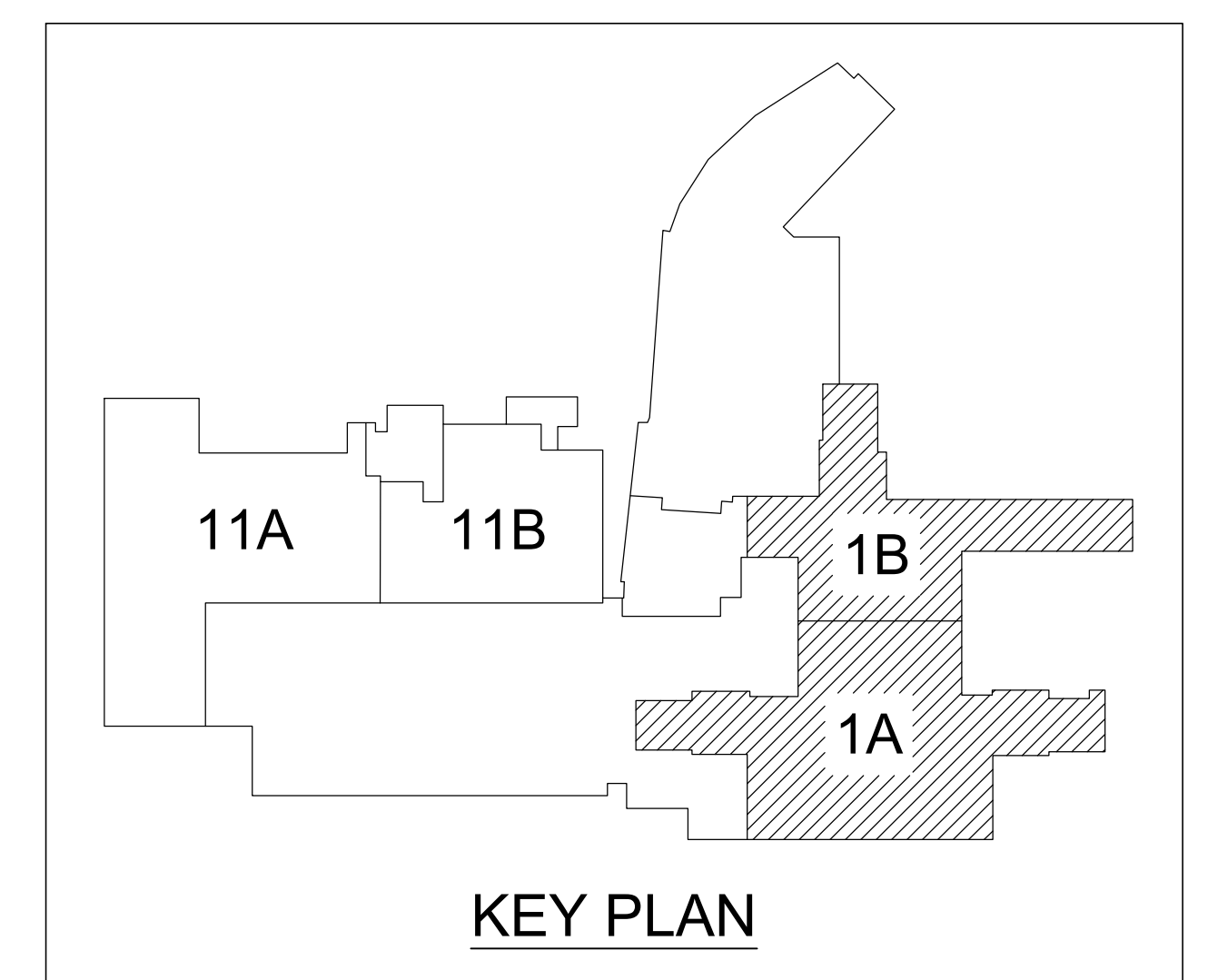


BUILDING # 1A & 1B - SEVENTH FLOOR PLAN

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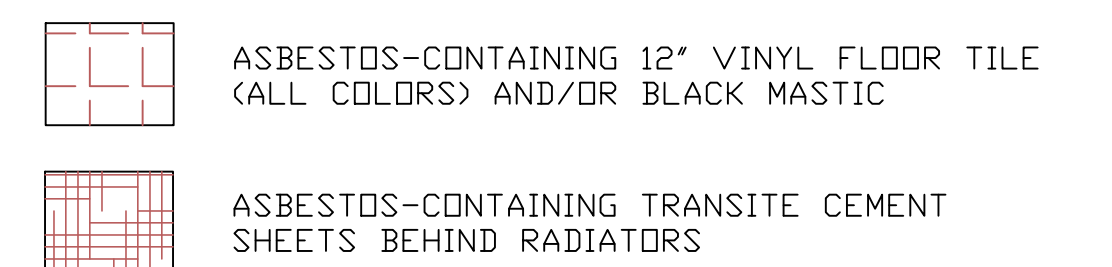
- ASBESTOS-CONTAINING 12" VINYL FLOOR TILE (ALL COLORS) AND/OR BLACK MASTIC
- ASBESTOS-CONTAINING TRANSITE CEMENT SHEETS BEHIND RADIATORS

Revisions	Dates	ENVIRONMENTAL CONSULTANT: CASTLE HILL ASSOCIATES, LLC. 218-A MECHANIC STREET, WATERVILLE, OHIO 43566 Tel: (567) 952-0212	 CASTLE HILL ASSOCIATES	Professional Seal: eV6	North: 	Drawing Title ARCHITECTURAL PLAN BUILDING 001 Scale: 1/16" = 1'-0"	Project Title FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS	Date SEPTEMBER 2019	 Office of Facilities	Department of Veterans Affairs		
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						Approved: Service Director	Location CHICAGO, ILLINOIS				Drawing No. 8A	
											Sheet 09 of 24	

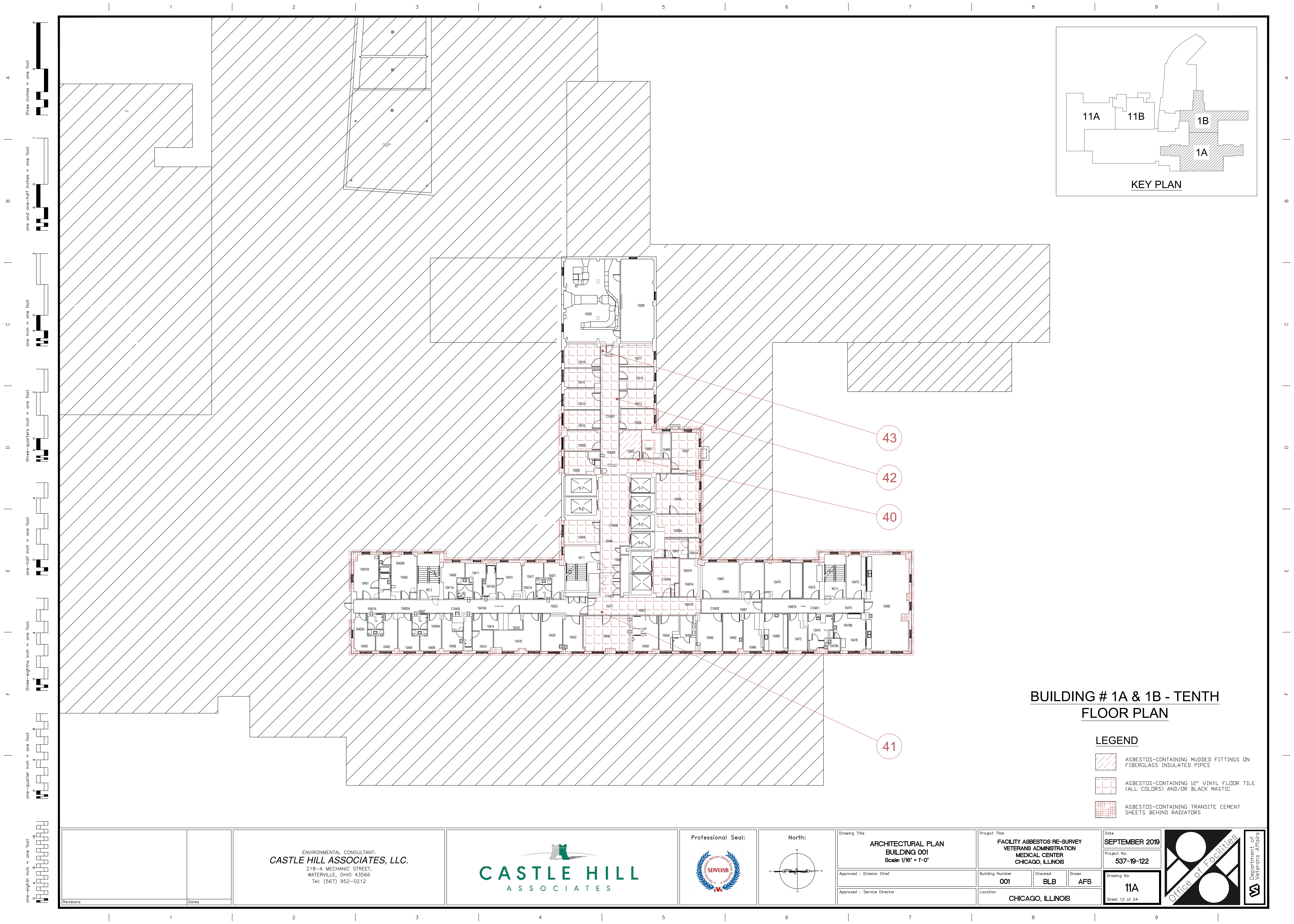


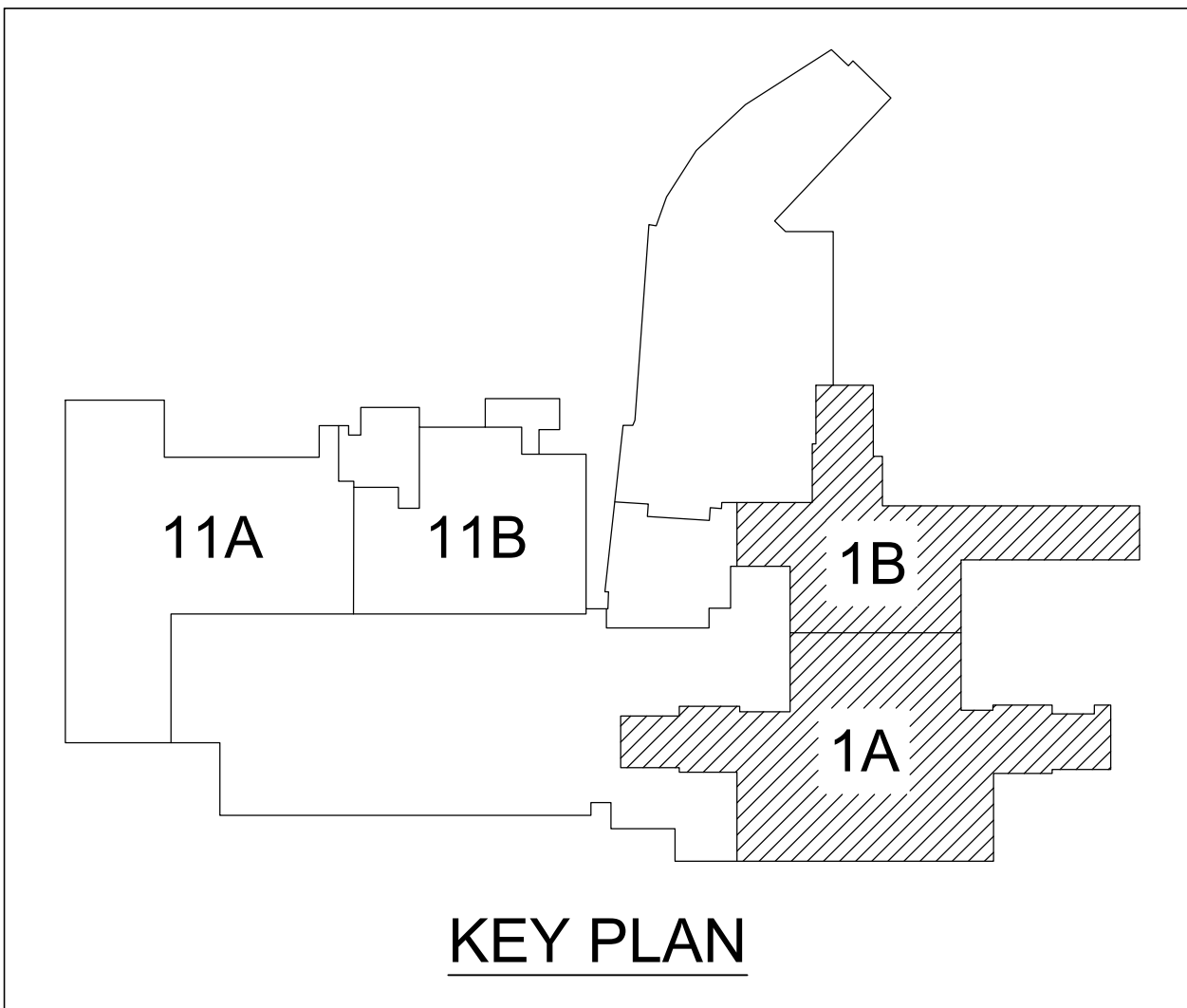
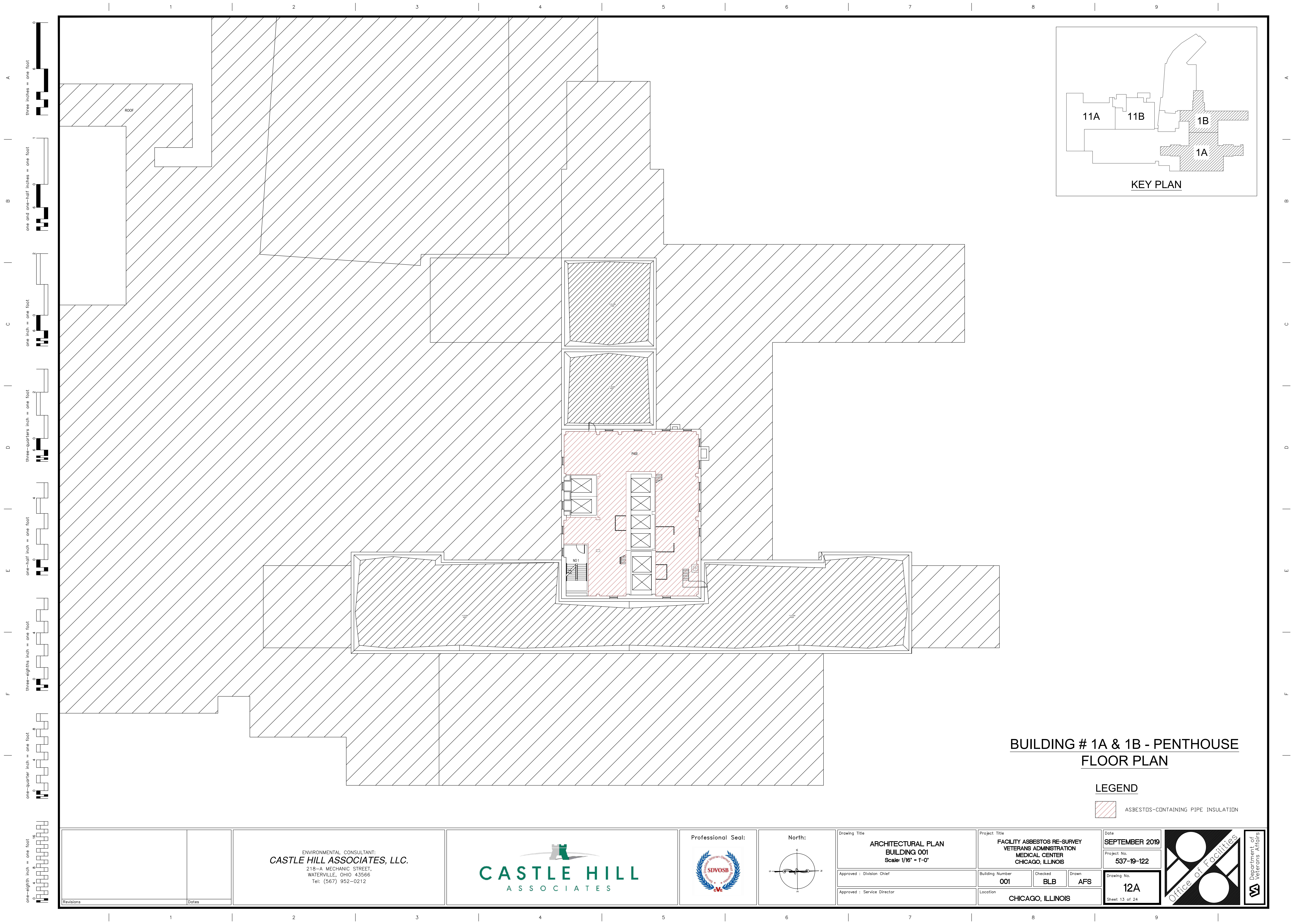
BUILDING # 1A & 1B - NINTH FLOOR PLAN

LEGEND



		ENVIRONMENTAL CONSULTANT: CASTLE HILL ASSOCIATES, LLC. 218-A MECHANIC STREET, WATERVILLE, OHIO 43566 Tel: (567) 952-0212		 CASTLE HILL ASSOCIATES		Professional Seal: 	North: 	Drawing Title ARCHITECTURAL PLAN BUILDING 001 Scale: 1/16" = 1'-0"	Project Title FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS	Date SEPTEMBER 2019 Project No. 537-19-122	
Revisions	Dates							Approved : Division Chief Approved : Service Director	Building Number 001 Checked BLB Drawn AFS	Drawing No. 10A Sheet 11 of 24	

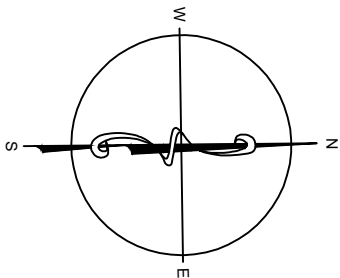


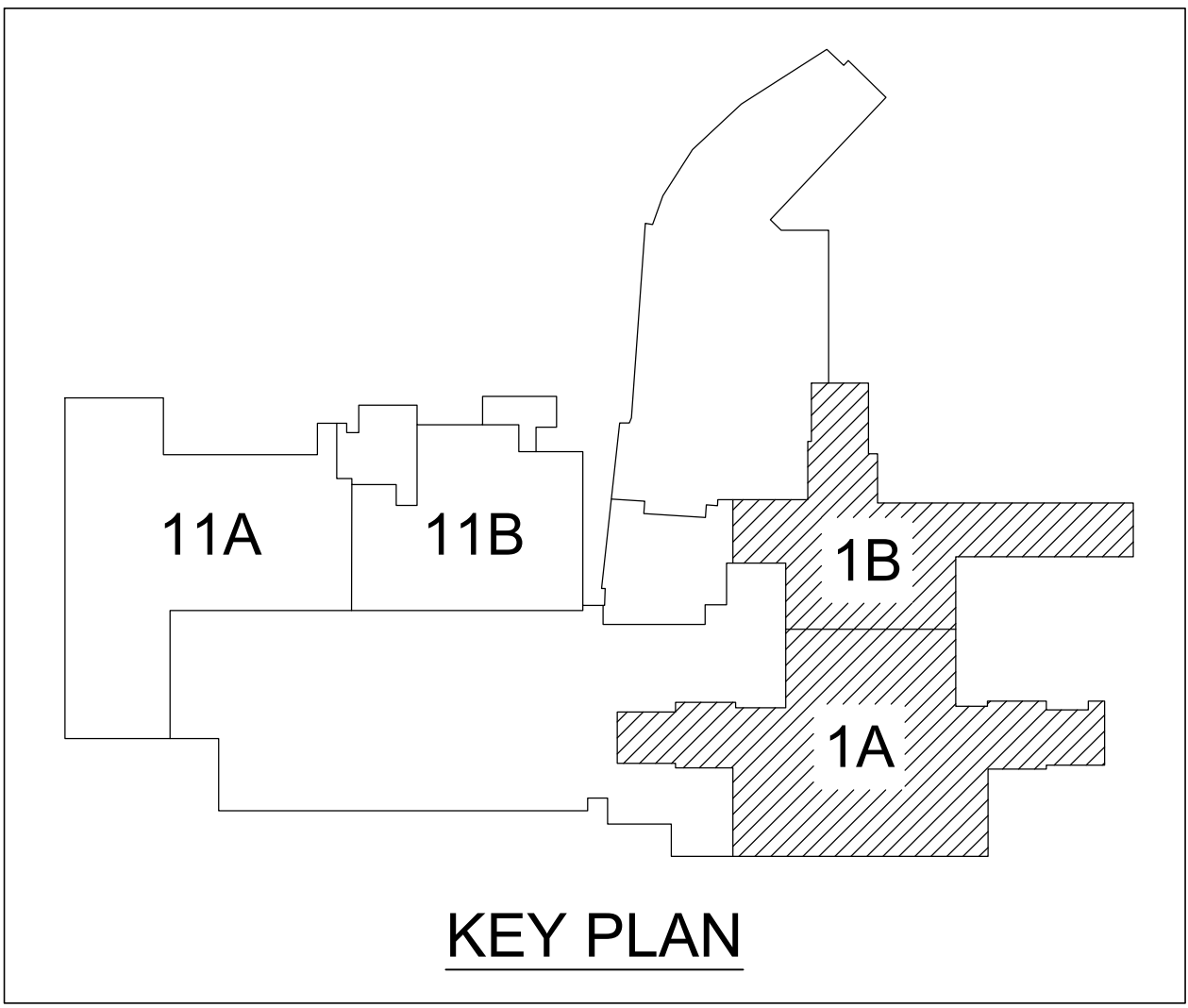
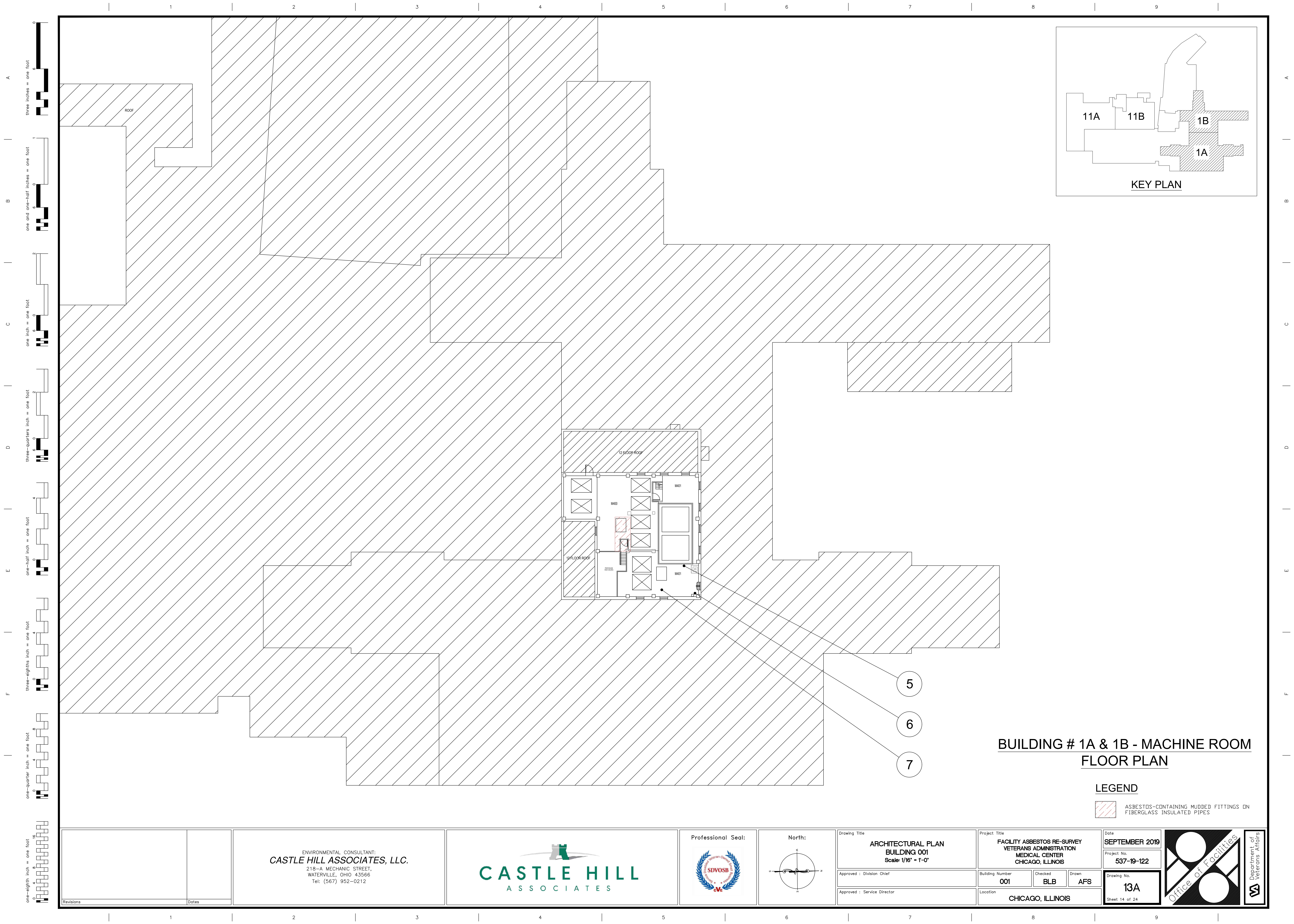


BUILDING # 1A & 1B - PENTHOUSE
FLOOR PLAN

LEGEND

ASBESTOS-CONTAINING PIPE INSULATION

Revisions	Dates	ENVIRONMENTAL CONSULTANT: CASTLE HILL ASSOCIATES, LLC. 218-A MECHANIC STREET, WATERVILLE, OHIO 43566 Tel: (567) 952-0212	 CASTLE HILL ASSOCIATES	Professional Seal: 	North: 	Drawing Title ARCHITECTURAL PLAN BUILDING 001 Scale: 1/16" = 1'-0"	Project Title FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS	Date SEPTEMBER 2019	 Office of Facilities	Department of Veterans Affairs 
						Approved : Division Chief	Building Number 001	Checked BLB		
						Approved : Service Director	Location CHICAGO, ILLINOIS		Sheet 13 of 24	

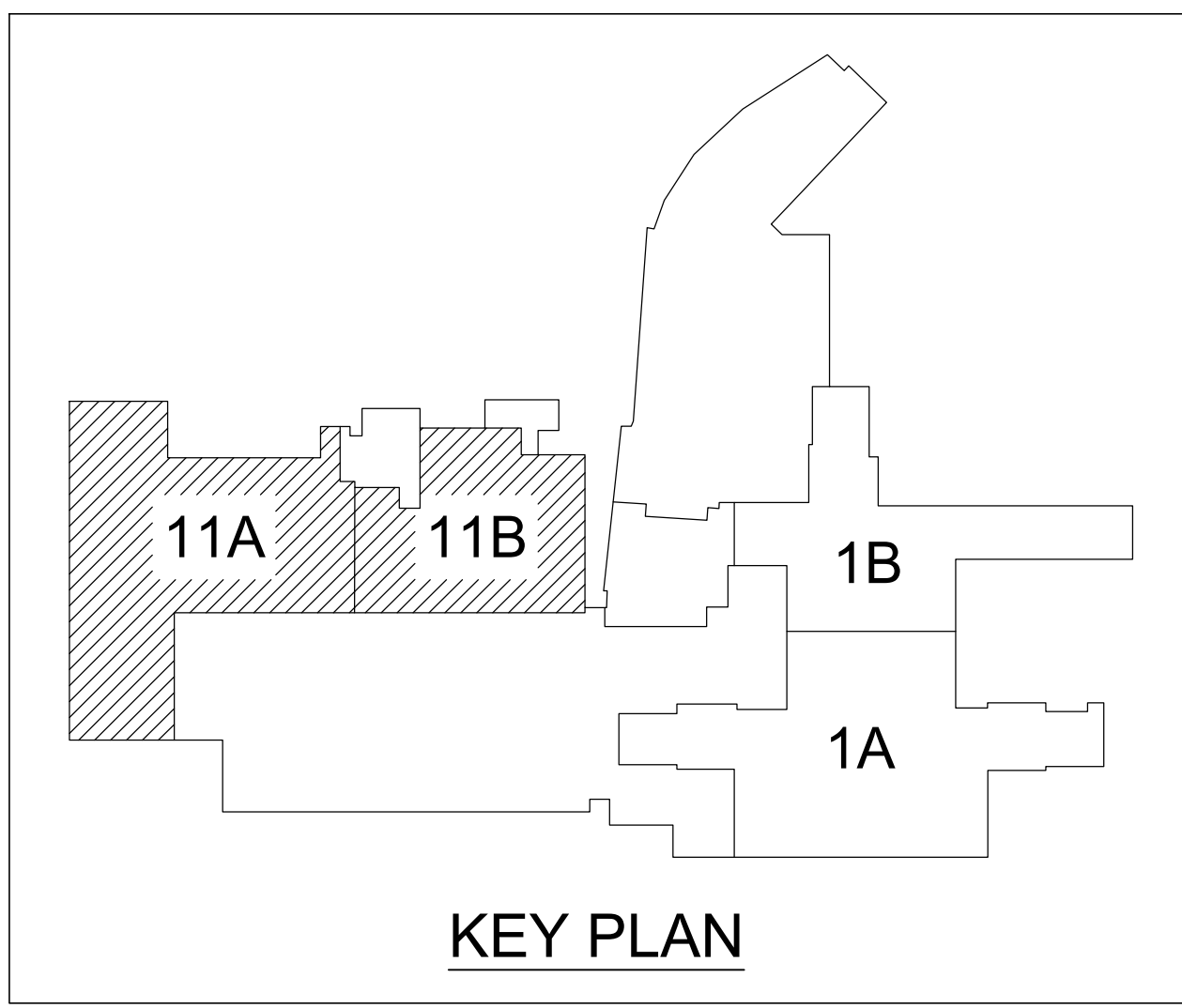


BUILDING # 1A & 1B - MACHINE ROOM
FLOOR PLAN

LEGEND

ASBESTOS-CONTAINING MUDDED FITTINGS ON
FIBERGLASS INSULATED PIPES

Revisions Dates		ENVIRONMENTAL CONSULTANT: CASTLE HILL ASSOCIATES, LLC. 218-A MECHANIC STREET, WATERVILLE, OHIO 43566 Tel: (567) 952-0212	CASTLE HILL ASSOCIATES	Professional Seal: 	North: 	Drawing Title ARCHITECTURAL PLAN BUILDING 001 Scale: 1/16" = 1'-0" Approved : Division Chief Approved : Service Director	Project Title FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS Building Number 001 Location CHICAGO, ILLINOIS	Date SEPTEMBER 2019 Project No. 537-19-122 Drawing No. 13A Sheet 14 of 24
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**BUILDING # 11A & 11B
SECOND FLOOR PLAN**

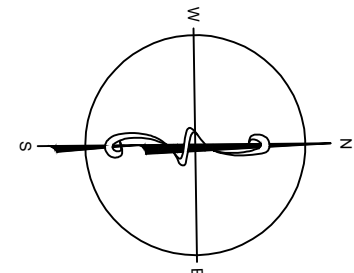
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CASTLE HILL ASSOCIATES, LLC.
218-A MECHANIC STREET,
WATERVILLE, OHIO 43566
Tel: (567) 952-0212



Professional Seal:



North:



Drawing Title
**ARCHITECTURAL PLAN
BUILDING 011
Scale: 1/16" = 1'-0"**

Approved : Division Chief

Approved : Service Director

Project Title
**FACILITY ASBESTOS RE-SURVEY
VETERANS ADMINISTRATION
MEDICAL CENTER
CHICAGO, ILLINOIS**

Building Number
011

Location
CHICAGO, ILLINOIS

Checked
BLB

Drawn
AFS

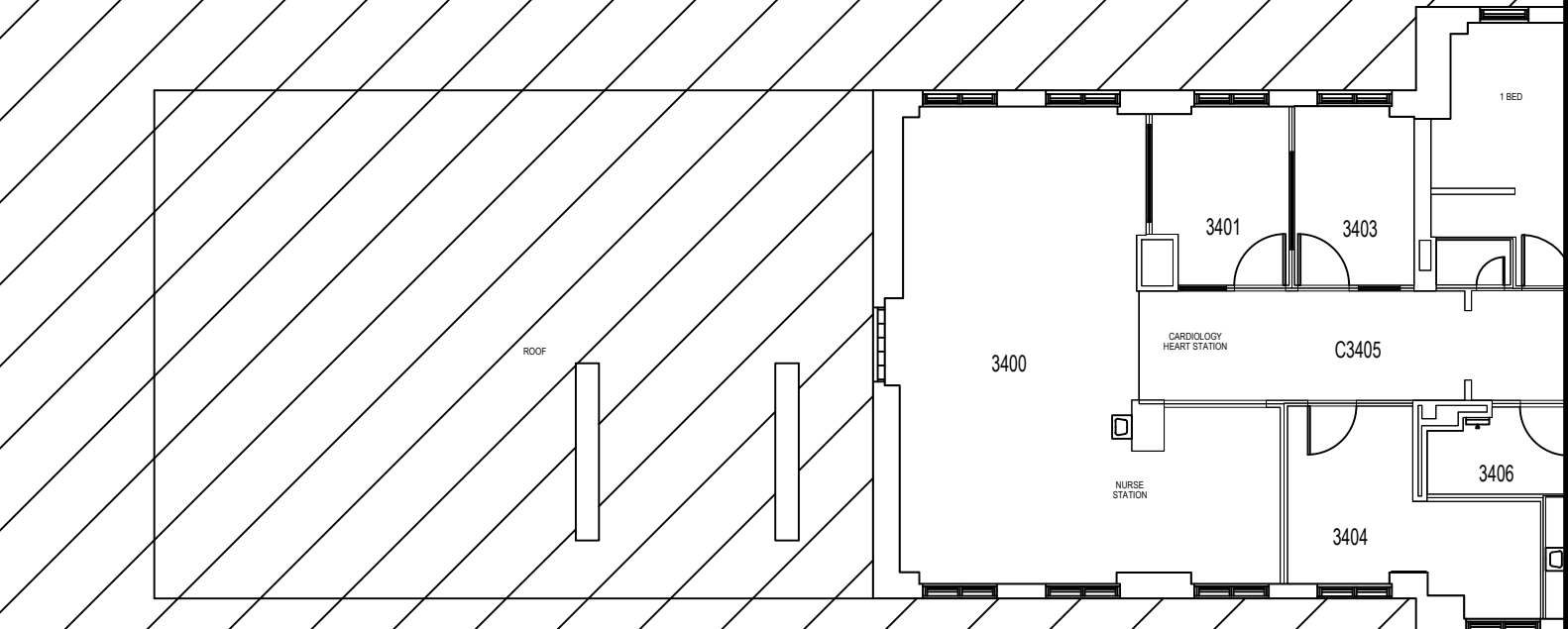
Date
SEPTEMBER 2019

Project No.
537-19-122

Drawing No.
3A

Sheet 17 of 24





THIRD FLOOR PLAN

LEGEND

 ASBESTOS-CONTAINING 12" VINYL FLOOR TILE (ALL COLORS)
AND/OR BLACK MASTIC

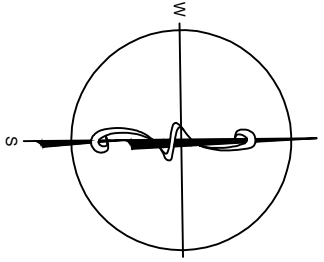
ENVIRONMENTAL CONSULTANT:
CASTLE HILL ASSOCIATES, LLC.
218-A MECHANIC STREET,
WATERVILLE, OHIO 43566
Tel: (567) 952-0212



Professional Seal



North



Drawing Title

ARCHITECTURAL PLAN
BUILDING 011
Scale: 1/16" = 1'-0"

Approved : Division Chief

Approved : Service Director

Project Title

FACILITY ASBESTOS RE-SURVEY
VETERANS ADMINISTRATION
MEDICAL CENTER
CHICAGO, ILLINOIS

Building Number
011

Location	CHICAGO, ILLINOIS
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Date

SEPTEMBER 2019

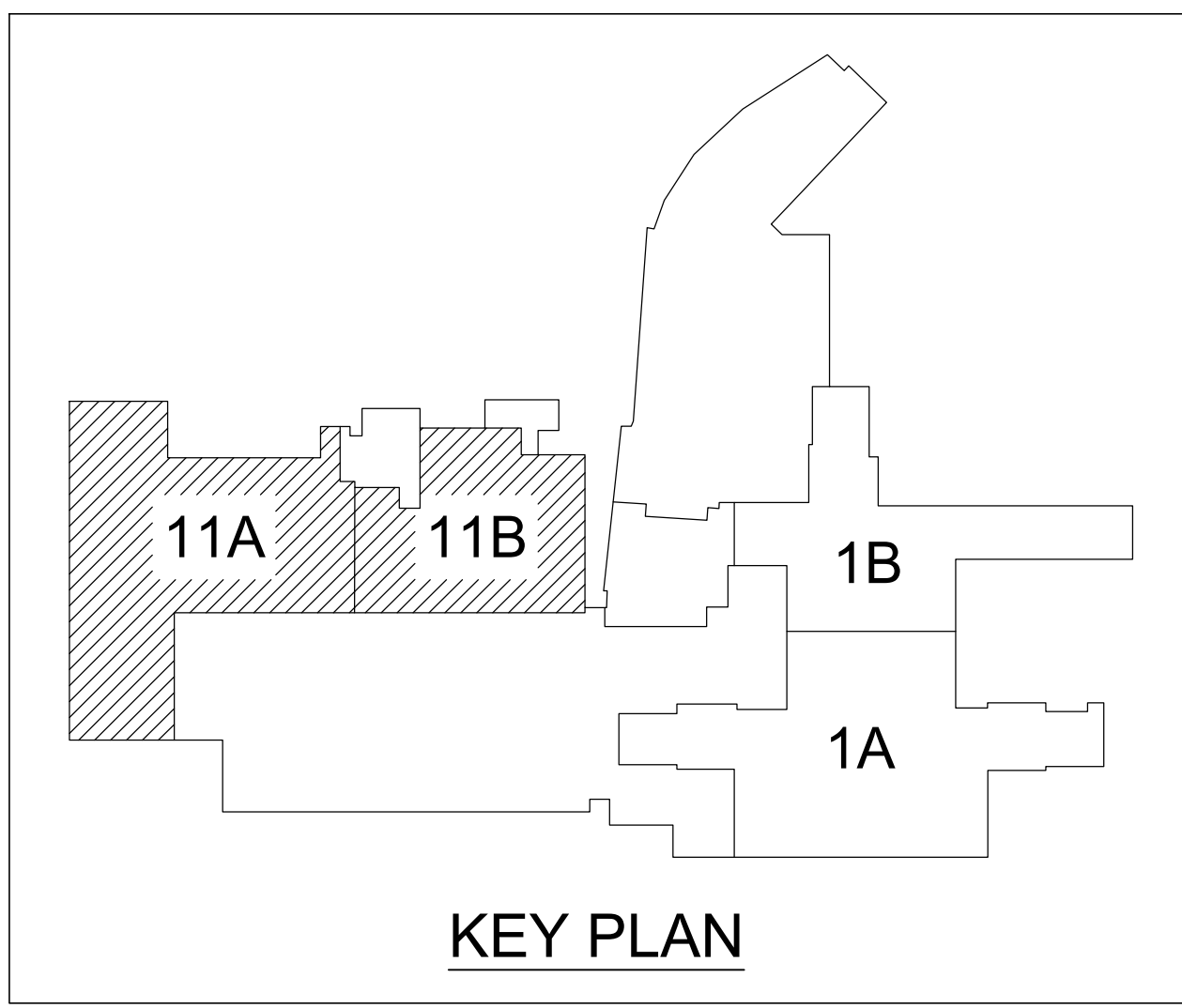
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Drawing No. 4

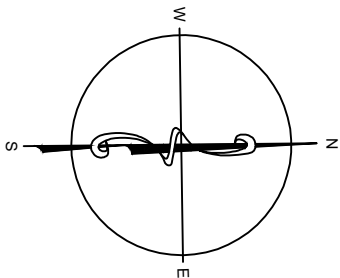
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BUILDING # 11A & 11B
FOURTH FLOOR PLAN

Revisions Dates		ENVIRONMENTAL CONSULTANT: CASTLE HILL ASSOCIATES, LLC. 218-A MECHANIC STREET, WATERVILLE, OHIO 43566 Tel: (567) 952-0212	 CASTLE HILL ASSOCIATES	Professional Seal: 	North: 	Drawing Title ARCHITECTURAL PLAN BUILDING 011 Scale: 1/16" = 1'-0" Approved : Division Chief Approved : Service Director	Project Title FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS Building Number 011 Location CHICAGO, ILLINOIS Checked BLB Drawn AFS	Date SEPTEMBER 2019 Project No. 537-19-122 Drawing No. 5A Sheet 19 of 24	 Office of Facilities  Department of Veterans Affairs
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BUILDING # 11A & 11B FIFTH FLOOR PLAN



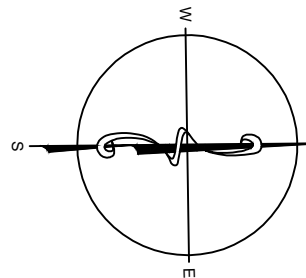
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CASTLE HILL ASSOCIATES, LLC.
218-A MECHANIC STREET,
WATERVILLE, OHIO 43566
Tel: (567) 952-0212



Professional Seal



North



Drawing Title

ARCHITECTURAL PLAN
BUILDING 011
Scale: 1/16" = 1'-0"

Approved : Division Chief

Approved : Service Director

Project Title

FACILITY ASBESTOS RE-SURVEY
VETERANS ADMINISTRATION
MEDICAL CENTER
CHICAGO, ILLINOIS

Building Number
011

Location

Date _____

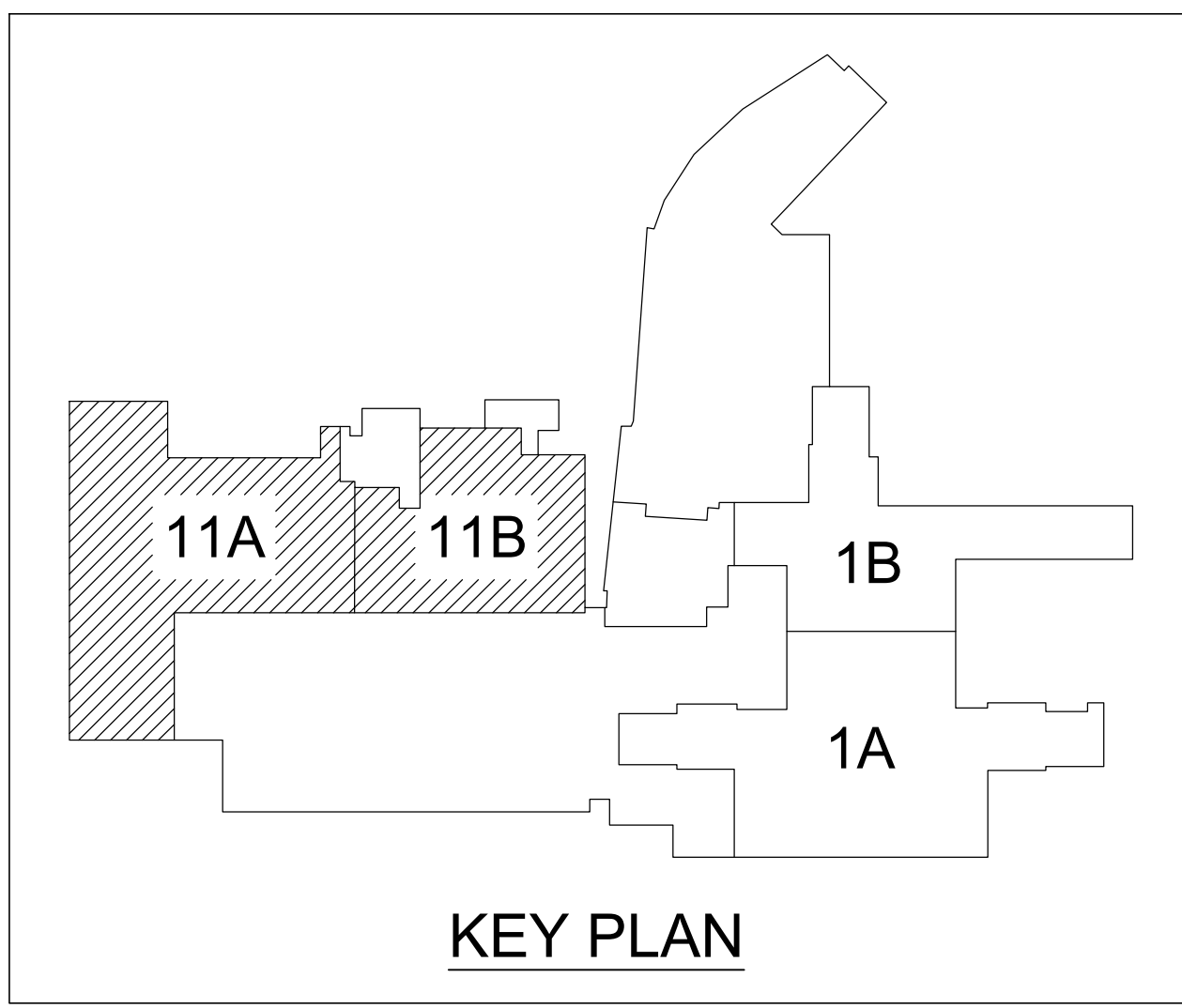
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Project No.
537-19-122

Drawing No.

Sheet 20 of 24

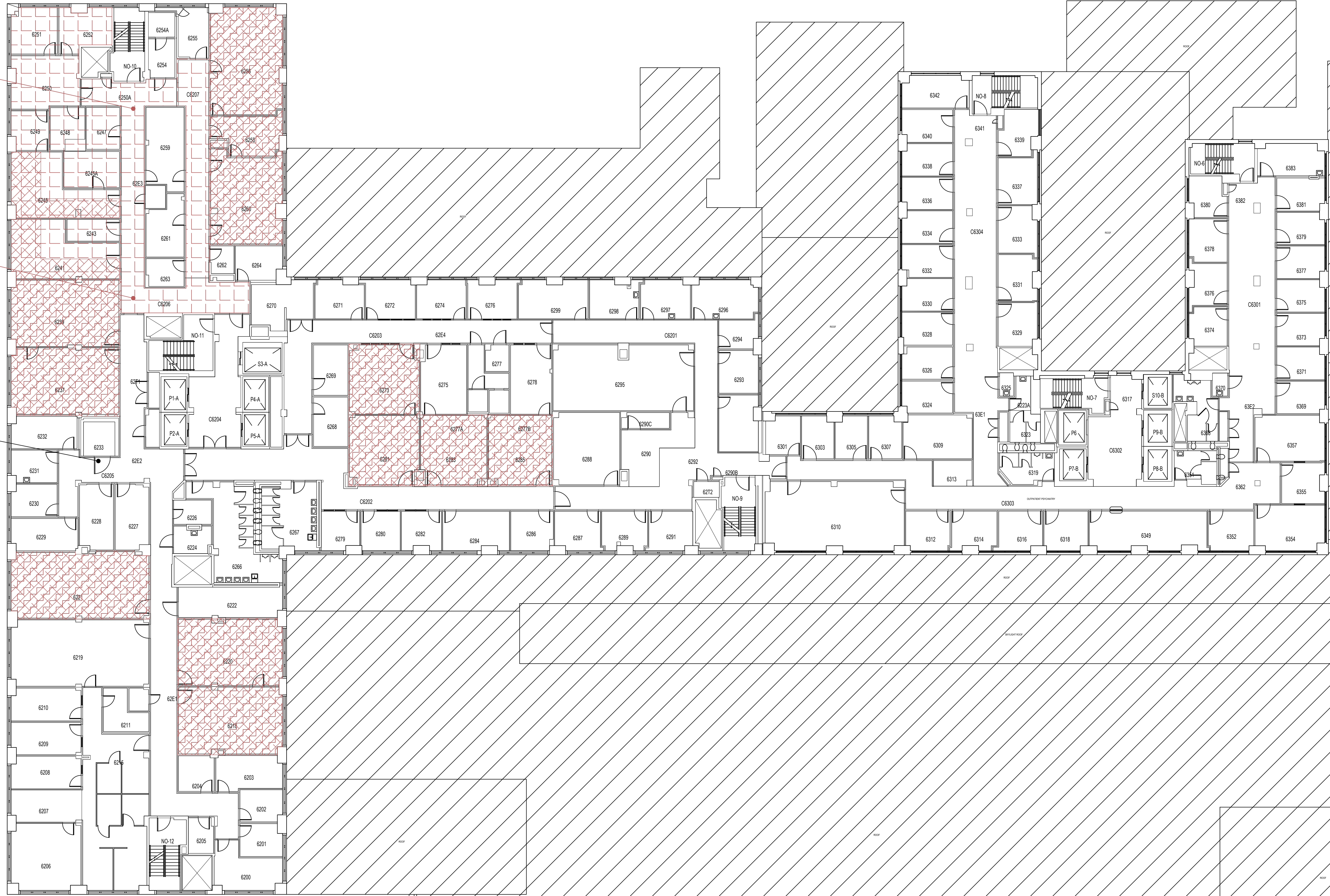




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BUILDING # 11A & 11B SIXTH FLOOR PLAN

LEGEND

- ASBESTOS-CONTAINING 12" VINYL FLOOR TILE (ALL COLORS) AND/OR BLACK MASTIC
- ASBESTOS-CONTAINING TRANSITE LABORATORY FUME HOODS
- ASBESTOS-CONTAINING EBONY TRANSITE LAB TABLE TOPS, SINKS, AND DRYING RACKS

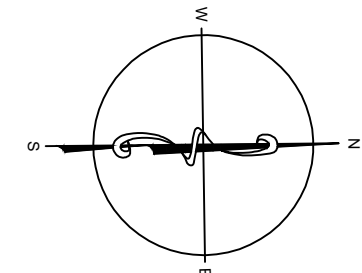
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CASTLE HILL ASSOCIATES, LLC.
218-A MECHANIC STREET,
WATERVILLE, OHIO 43566
Tel: (567) 952-0212



Professional Seal:



North:



Drawing Title
**ARCHITECTURAL PLAN
BUILDING 011
Scale: 1/16" = 1'-0"**

Approved : Division Chief

Approved : Service Director

Project Title
**FACILITY ASBESTOS RE-SURVEY
VETERANS ADMINISTRATION
MEDICAL CENTER
CHICAGO, ILLINOIS**

Building Number
011

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BLB

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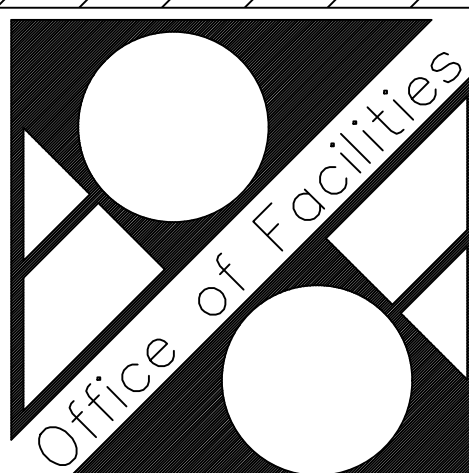
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Date
SEPTEMBER 2019

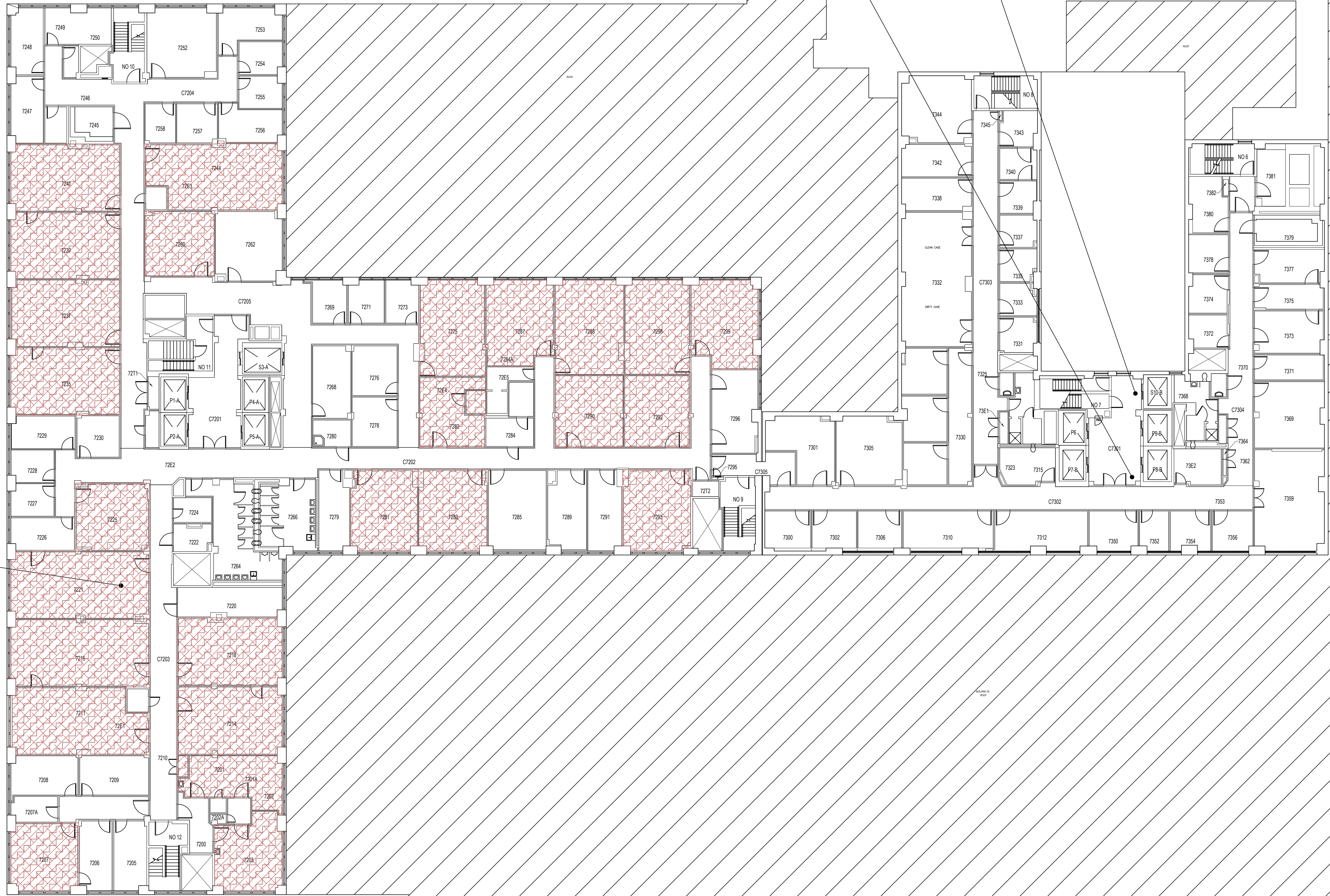
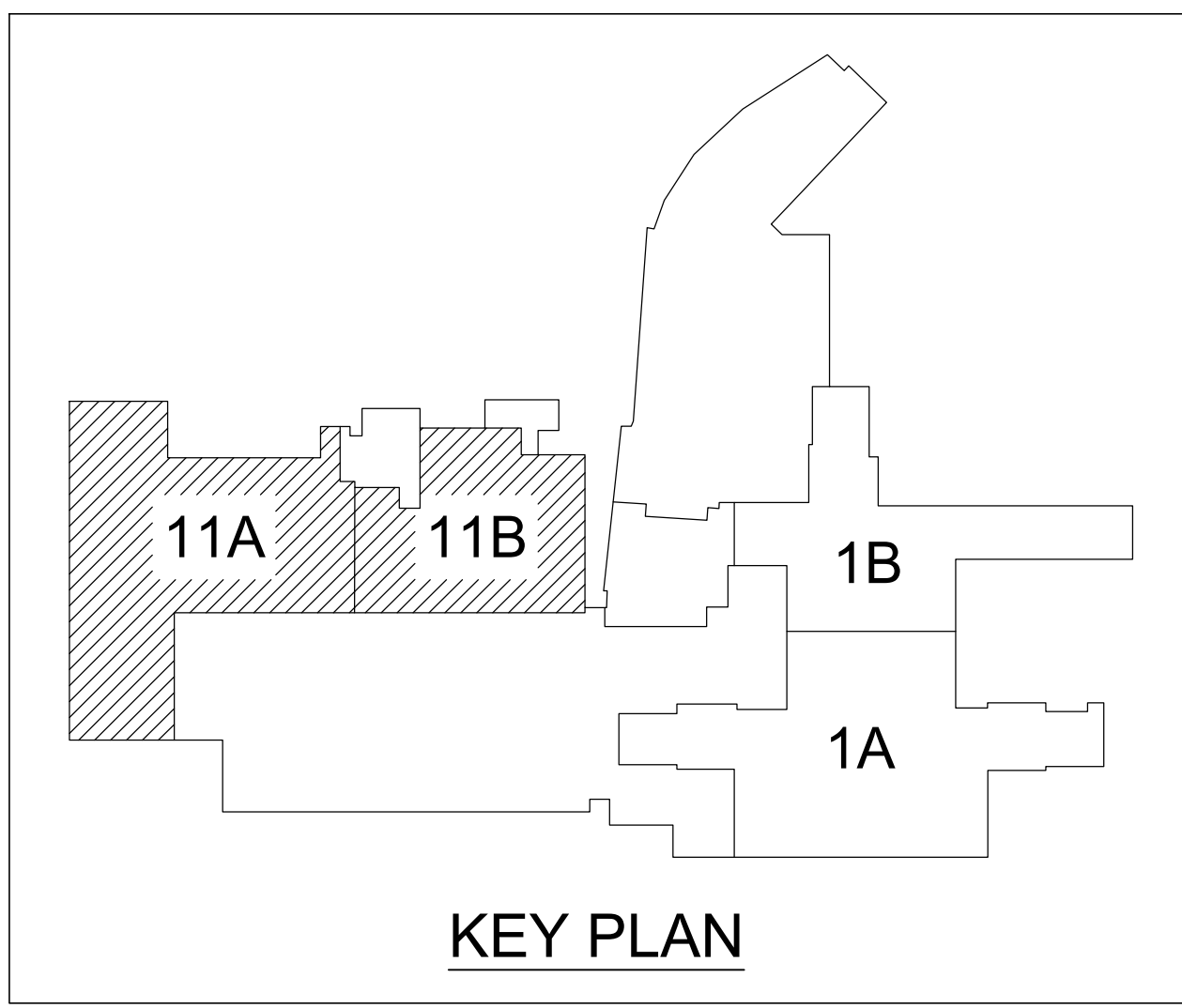
Project No.
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Drawing No.
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Sheet 21 of 24

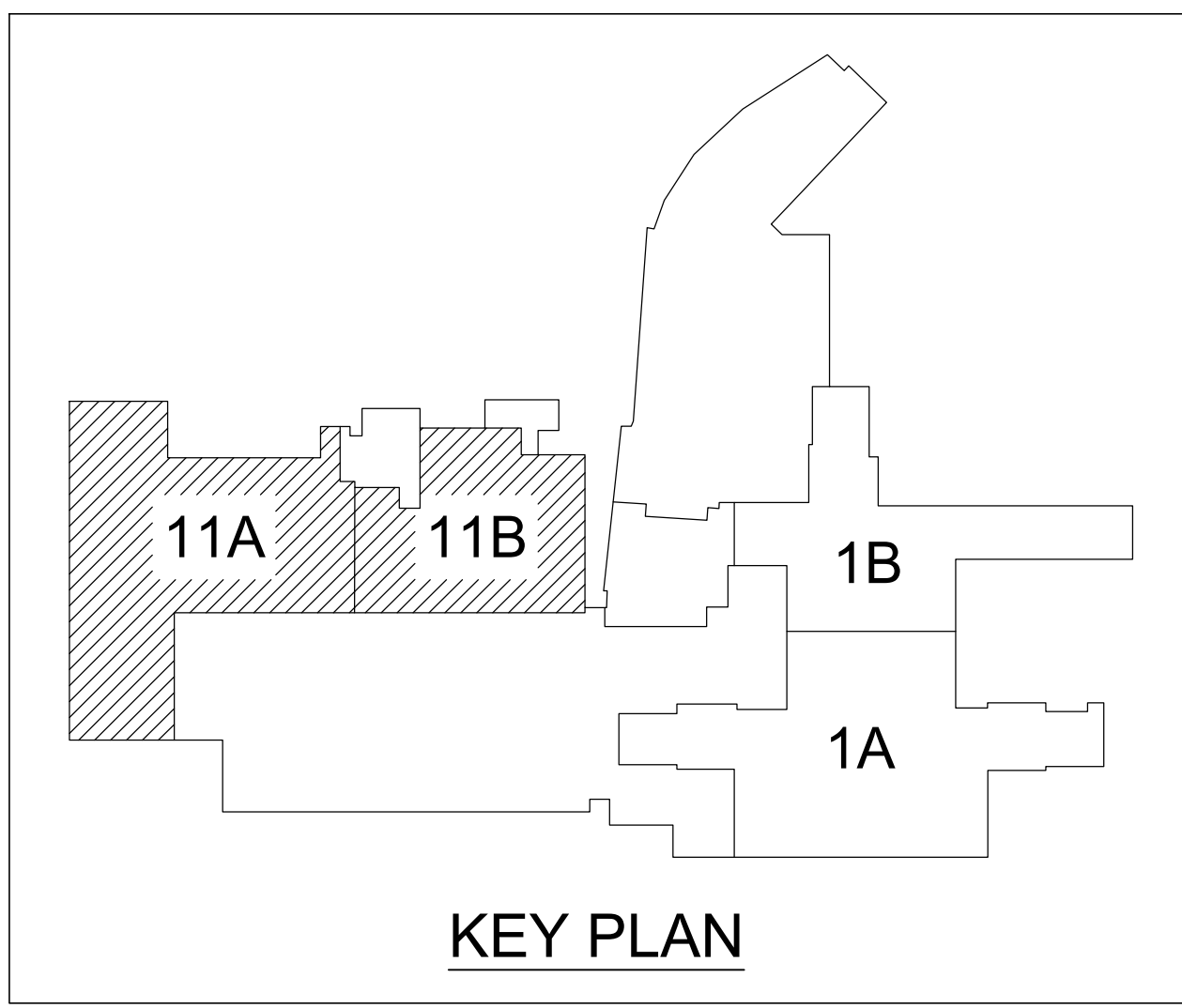


Department of
Veterans Affairs



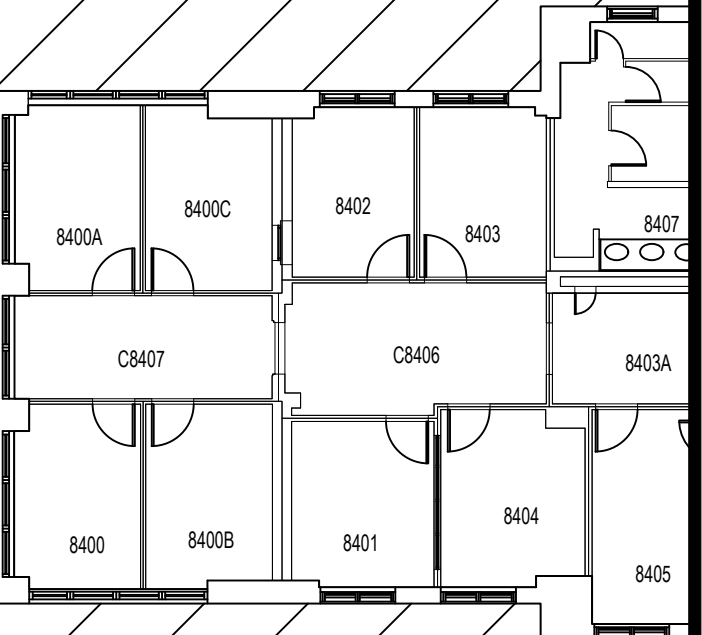
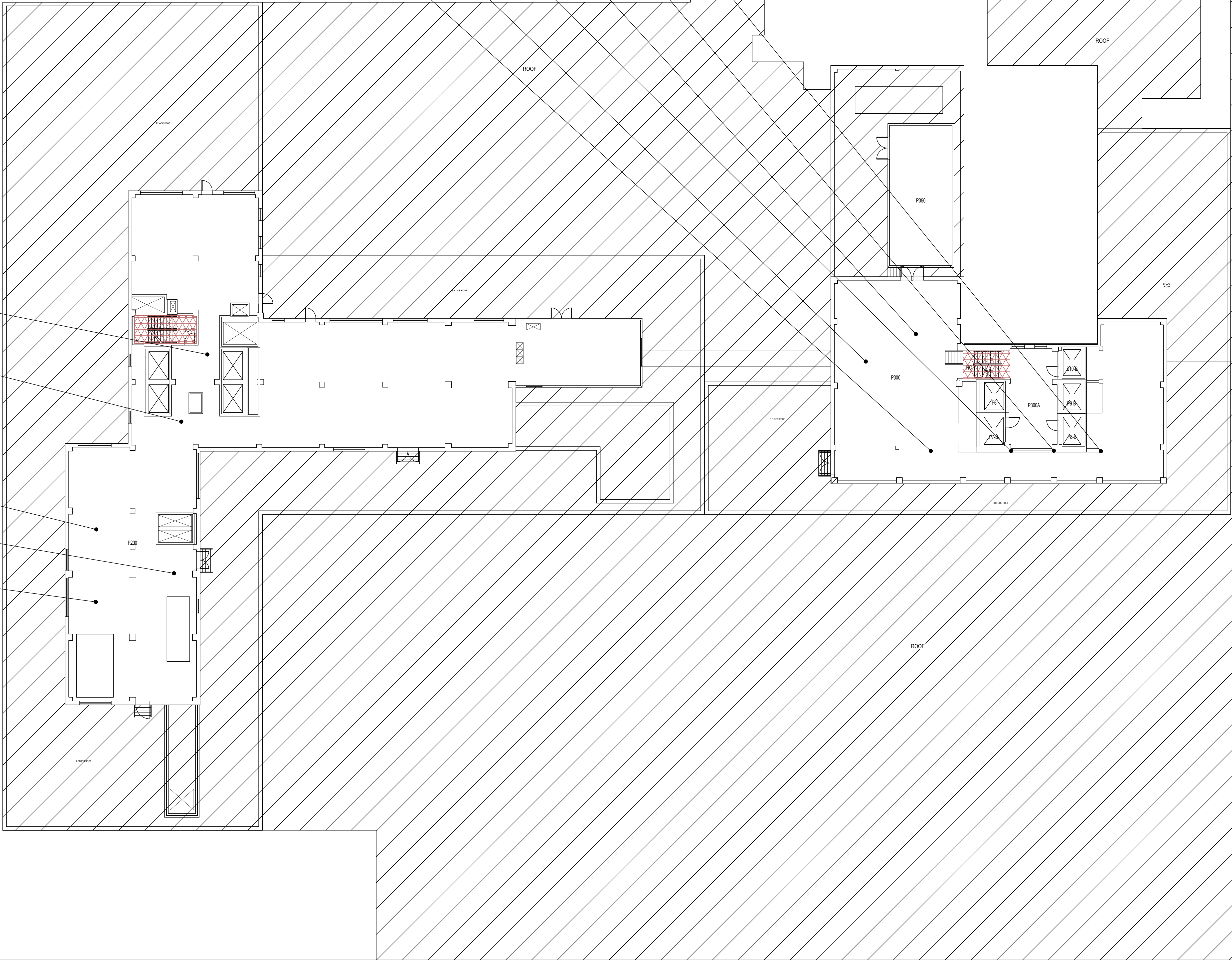
- LEGEND**
- ASBESTOS-CONTAINING TRANSITE LABORATORY FUME HOODS
 - ASBESTOS-CONTAINING EBONY TRANSITE LAB TABLE TOPS, SINKS, AND DRYING RACKS

Revisions	Dates	ENVIRONMENTAL CONSULTANT: CASTLE HILL ASSOCIATES, LLC. 218-A MECHANIC STREET, WATERVILLE, OHIO 43566 Tel: (567) 952-0212	 CASTLE HILL ASSOCIATES	Professional Seal: 	North: 	Drawing Title ARCHITECTURAL PLAN BUILDING 011 Scale: 1/16" = 1'-0" Approved : Division Chief Approved : Service Director	Project Title FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS Building Number 011 Location CHICAGO, ILLINOIS	Date SEPTEMBER 2019 Project No. 537-19-122 Drawing No. 8A Sheet 22 of 24	 Office of Facilities Department of Veterans Affairs
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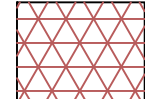
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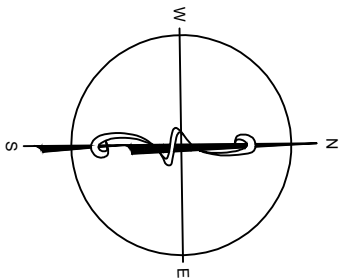
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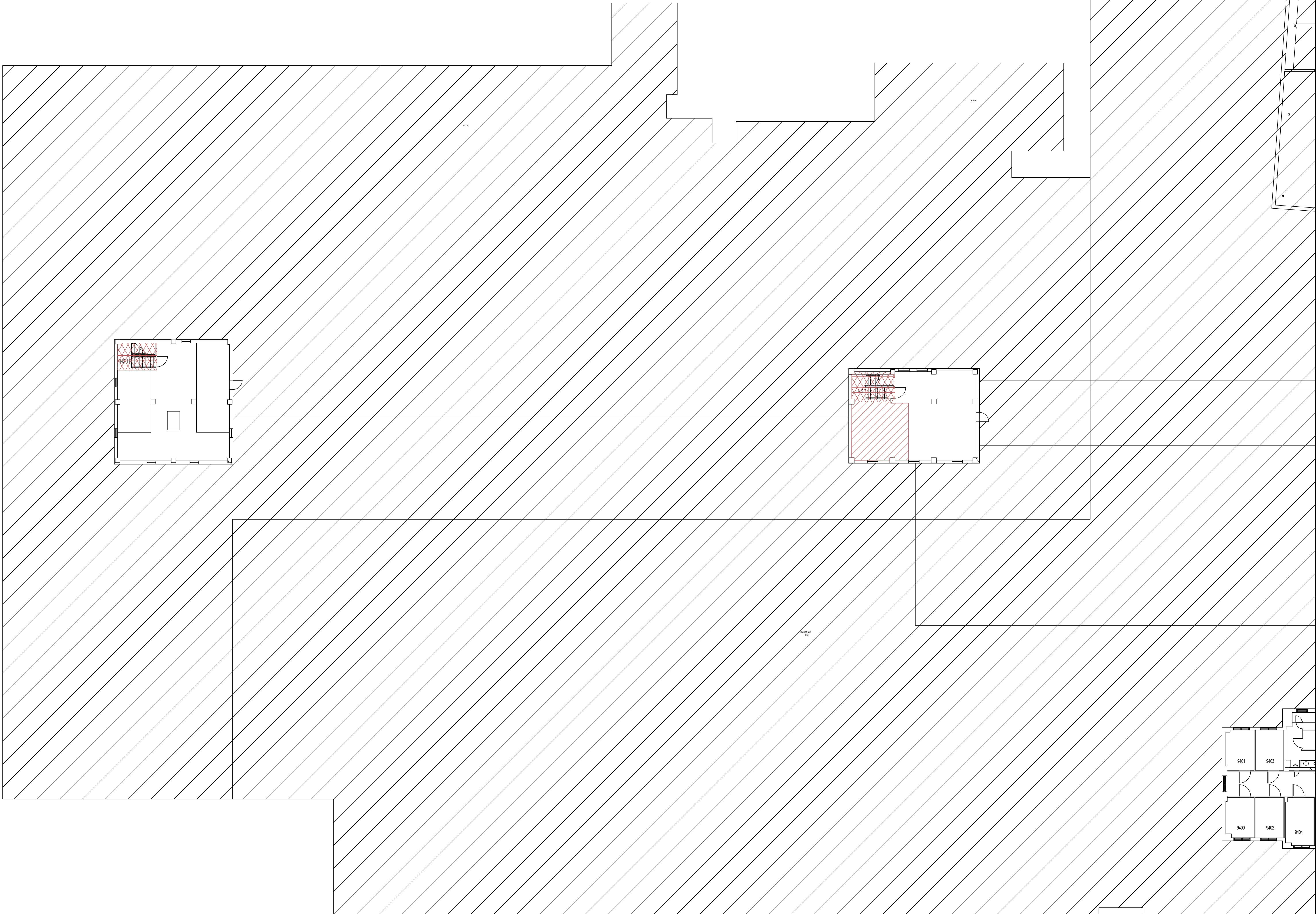
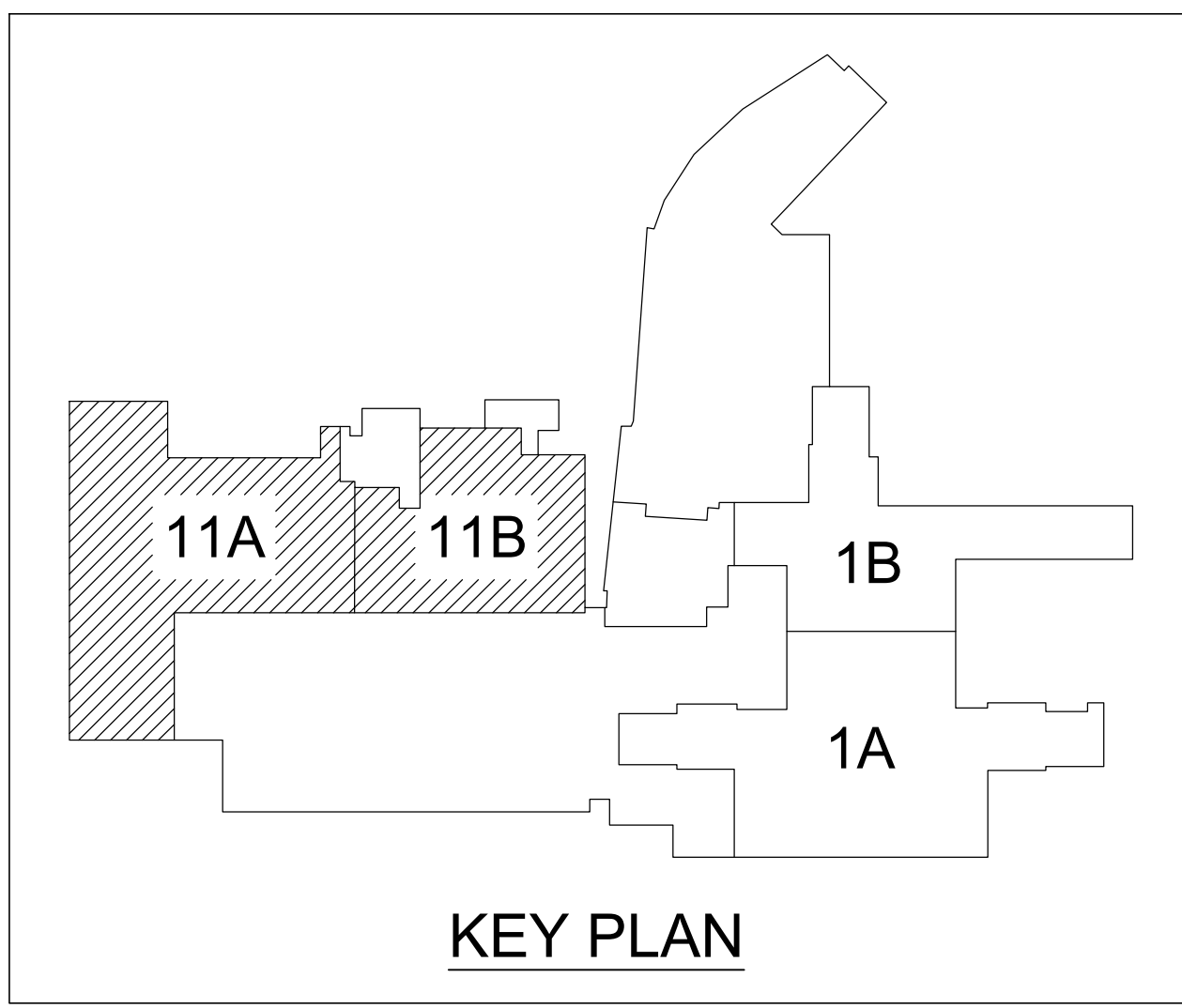


BUILDING # 11A & 11B PENTHOUSE FLOOR PLAN

LEGEND

 ASBESTOS-CONTAINING TRANSITE SHEETS IN STAIRWELLS

Revisions Dates	ENVIRONMENTAL CONSULTANT: CASTLE HILL ASSOCIATES, LLC. 218-A MECHANIC STREET, WATERVILLE, OHIO 43566 Tel: (567) 952-0212		Professional Seal: 	North: 	Drawing Title ARCHITECTURAL PLAN BUILDING 011 Scale: 1/16" = 1'-0" Approved : Division Chief Approved : Service Director	Project Title FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS Building Number 011 Location CHICAGO, ILLINOIS Checked BLB Drawn AFS	Date SEPTEMBER 2019 Project No. 537-19-122 Drawing No. 9A Sheet 23 of 24	
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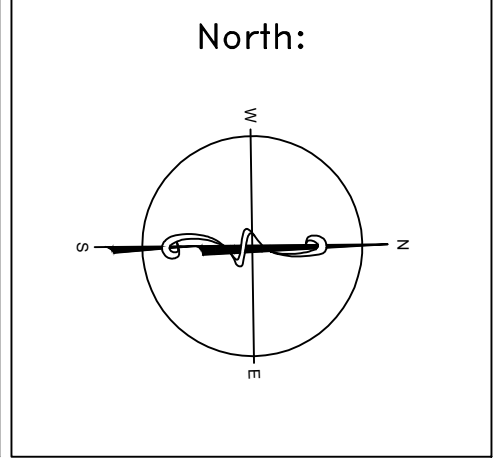


**BUILDING # 11A & 11B
MACHINE ROOM FLOOR PLAN**

- LEGEND**
- ASBESTOS-CONTAINING PIPE INSULATION
 - ASBESTOS-CONTAINING TRANSITE SHEETS IN STAIRWELLS

Revisions	Dates

ENVIRONMENTAL CONSULTANT:
CASTLE HILL ASSOCIATES, LLC.
218-A MECHANIC STREET,
WATERVILLE, OHIO 43566
Tel: (567) 952-0212



Drawing Title	ARCHITECTURAL PLAN BUILDING 011 Scale: 1/16" = 1'-0"
Approved : Division Chief	
Approved : Service Director	

Project Title	FACILITY ASBESTOS RE-SURVEY VETERANS ADMINISTRATION MEDICAL CENTER CHICAGO, ILLINOIS
Building Number	011
Checked	BLB
Drawn	AFS
Location	CHICAGO, ILLINOIS

Date	SEPTEMBER 2019
Project No.	537-19-122
Drawing No.	10A
Sheet	24 of 24



APPENDIX H

O&M PLAN

(OPERATIONS & MAINTENANCE)

JESSE BROWN VA MEDICAL CENTER ASBESTOS OPERATIONS AND MAINTENANCE PLAN

FY2019

OCTOBER 2019



**US DEPARTMENT OF VETERAN AFFAIRS
JESSE BROWN VA MEDICAL CENTER
CHICAGO, IL**

STANDARD OPERATING PROCEDURES

**PREPARED BY:
CASTLE HILL ASSOCIATES, LLC
218-A MECHANIC ST.
WATERVILLE, OH 43566**



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1.0 Introduction

Asbestos is a common building material used in the United States from the early 1900's to present. Asbestos is completely safe when maintained in "good" condition, but becomes a potential hazard if the microscopic fibers are released into the air. This occurs when the material containing the asbestos becomes damaged and is friable-meaning when it is dry, it can be crumbled in your hand. If the dust containing the asbestos fibers is inhaled, these fibers can become trapped within the body and may eventually lead to asbestosis, mesothelioma, and lung cancer. If ingested, it may lead to many other types of cancer.

1.1 Regulatory History of Asbestos Bans

- In 1973, EPA banned spray-applied surfacing asbestos-containing material for fireproofing/insulating purposes. See National Emission Standards for Hazardous Air Pollutants (NESHAP) at [40 CFR Part 61, Subpart M](#).
- In 1975, EPA banned installation of asbestos pipe insulation and asbestos block insulation on facility components, such as boilers and hot water tanks, if the materials are either pre-formed (molded) and friable or wet-applied and friable after drying. See National Emission Standards for Hazardous Air Pollutants (NESHAP) at [40 CFR Part 61, Subpart M](#).
- In 1978, EPA banned spray-applied surfacing materials for purposes not already banned. See National Emission Standards for Hazardous Air Pollutants (NESHAP) at [40 CFR Part 61, Subpart M](#).
- In 1977, the [Consumer Product Safety Commission \(CPSC\)](#) banned the use of asbestos in artificial fireplace embers and wall patching compounds. ([16 CFR Part 1305](#) and [16 CFR 1304](#)).
- In 1989, the EPA issued a final rule under Section 6 of Toxic Substances Control Act (TSCA) banning most asbestos-containing products. However, in 1991, this rule was vacated and remanded by the Fifth Circuit Court of Appeals. As a result, most of the original ban on the manufacture, importation, processing, or distribution in commerce for the majority of the asbestos-containing products originally covered in the 1989 final rule was overturned. See [40 CFR 763 Subpart I](#).

2.0 Purpose

The purpose of the Asbestos Operations and Maintenance (O&M) Plan is to address the existing Asbestos Containing Materials (ACM)/Asbestos Containing Building Materials (ACBM) within the Jesse Brown VA Medical Center facilities and how to prevent/minimize the potential for asbestos fiber exposure to building occupants, visitors, residents, employees, etc. This plan will identify procedures to maintain any existing ACM/ACBM in 'good' condition until all ACM/ACBM is ultimately abated. All abatement activities conducted at the Jesse Brown VAMC will be conducted by licensed abatement contractors and no abatement activities will be conducted by VA personnel. This plan will include the updated details of routine surveys and inspections and the procedures for handling ACM/ACBM that may become 'damaged' or 'significantly damaged'.

The following definitions are the different conditions for all ACM/ACBM and how to identify what condition the ACM/ACBM is in:

1. Good: <1% overall damage and any damage does not appear to be a significant hazard.
2. Damaged: <10% localized damage or <25% total damage throughout homogeneous area.

3. Significantly Damaged: >10% localized damage or >25% total damage throughout homogeneous area.

Anytime a condition other than 'good' is identified, it shall be reported to the Asbestos Program Manager immediately.

The following OSHA asbestos regulations are applicable and must be followed where appropriate:

1. 29 CFR 1910.1001 General Industry:
 - a. Employees must not be exposed to asbestos fibers at or above the permissible exposure level (PEL) as per **29 CFR 1910.1001I**.
 - b. Employees must be notified of ACM/ACBM in the area as per **29 CFR 1910.1001(j)**.
 - c. If exposure limit reaches the PEL, proper procedures must be followed as per **29CFR 1910.1001I and (f)**.
 - d. If entry into this area is required, then **29 CFR 1910.1001(g), (h), (i), and (l)** must be followed.
2. 29 CFR 1926.1101 Construction:
 - a. Pertains to construction projects, repair operations, etc. as defined in **29 CFR 1910.12(b)**. Also includes housekeeping during and right after construction.
 - b. There are different procedures according to the class of asbestos work as defined in **29 CFR 1926.1101 'ASBESTOS'**; Class I, II, III, and IV which must be followed. See below.
 - c. If work is class I, II, or III, then the area must be regulated as per **29 CFR 1926.1101I**.
 - d. Regardless of the level of exposure of asbestos fibers, engineering controls and work practices must be taken, as per **29 CFR 1926.1101 (g)** and the specific class of asbestos work performed.
 - e. Proper notifications must be made during all asbestos-related work as per **29 CFR 1926.1101(k)**.

The different classifications of asbestos work are:

1. Class I: Activities involving the removal of TSI and surfacing ACM and PACM.
2. Class II: Activities involving the removal of ACM which is not thermal system insulation or surfacing material. This includes, but is not limited to, the removal of asbestos-containing wallboard, floor tile and sheeting, roofing and siding shingles, and construction mastics.
3. Class III: Repair and maintenance operations, where "ACM", including TSI and surfacing ACM and PACM, is likely to be disturbed (conducted by outside abatement contractors).
4. Class IV: Maintenance and custodial activities during which employees contact but do not disturb ACM or PACM and activities to clean up dust, waste and debris resulting from Class I, II, and III activities (this is only class of work which will be conducted by VA employees).

2.1 Scope and Objectives

The primary goal for the Asbestos Operation and Maintenance Plan is to prevent the release of asbestos fibers from existing ACM/ACBM. If prevention is not possible, then the O&M Plan will minimize the exposure of asbestos fibers. The O&M Plan will accomplish this by:

- Maintaining existing ACM/ACBM in 'good' condition.
- Routinely inspect the condition of known ACM/ACBM in a safe manner – without disturbing the material.
- Establishing procedures for managing ACM/ACBM that becomes damaged or has a potential for damage/significant damage.
- Respond to emergencies involving the disturbance of asbestos.
- Keep a record of any surveys, inspections, CAD drawings, and document changes in the existence/condition of all ACM/ACBM, any newly discovered ACM/ACBM, and any abatement of ACM/ACBM. These records will be located in the following location:

Michael C. Schiro
Engineering Department - 3rd Floor
Office: 312-569-8599
Cell 312-523-1665
Michael.Schiro2@va.gov

3.0 Roles and Responsibilities

This section will identify the roles and responsibilities:

1. Facilities Management:

- a. *Chief, Facilities Management Services* – Manage, maintain and implement the Asbestos Management Program and O&M Plan.
- b. *Chief, Maintenance and Operations* – Directs Operations and Maintenance staff to contract all abatement work in compliance with the plan, report any deficiencies, newly discovered ACM, and newly damaged ACM, etc.
- c. *Asbestos Program Manager* – Performs routine updates to the plan and files relevant documentation.

2. Safety Section:

- a. *Safety Manager* – Provides guidance and support to ensure environmental program compliance.
- b. *Industrial Hygienist/GEMS Manager* – Performs 6-month surveillance and ACM Sampling as needed.

3. VISN 6 Contracting Office:

Provide government acquisition support for contracted abatement services.

4.0 Communication

This section describes the steps used to notify building occupants, contractors, and employees of the presence of asbestos in the building, abatement activities, and warning labels and caution signs. Good communication between management, tenants, contractors, and employees ensures:

- All regulatory requirements are met;
- All abatement work is performed by licensed abatement contractors in accordance with the O&M Plan and applicable documents;
- VA personnel are prevented from disturbing asbestos; and
- Misunderstandings are prevented.

4.1 Communication with Patients, Visitors, and Employees

Being a healthcare facility, in some cases, it may be more practical to relocate patients, visitors and employees to another area within the facility versus formal notifications. For asbestos abatement projects, patients, visitors and employees will be relocated during the construction phase.

4.2 Outside Contractor Communication

When contractors perform asbestos abatement within the facility, the contract documents normally define the amount, type, and condition of all known ACM/ACBM to be abated prior to contract award. If undefined ACM/ACBM is encountered during construction, the contractor will immediately notify the construction project manager/COR. Existing ACM that is not disturbed by the construction work will not always be abated. Contractors are responsible for performing all contract defined ACM abatement work in accordance with applicable federal, state, and local regulations and laws.

4.3 Employee Communication

Abatement contractors are required to comply with the OSHA regulations [29CFR 1910.1001](#) for General Industry and OSHA regulations [29CFR 1926.1101](#) during construction. There are specific details outlined depending on employee's duties and exposure levels. More specifically, refer to:

1. [29CFR 1900.1001\(d\): 'Exposure Monitoring'; \(7\): 'Employee notification of monitoring results'](#)
2. [29CFR 1900.1001\(j\): 'Communication of hazards to employees – Introduction'](#)
3. [29CFR 1926.1101\(k\): 'Communication of hazards'](#)

4.4 Notification to Agencies

This section outlines the notification requirements for the State of Illinois and the Environmental Protection Agency (EPA) compliance. Depending on the type and amount of asbestos work, scheduled or emergency, the notification requirements will vary as follows.

4.4.1 Notification Requirements

Projects that do not disturb ACM/ACBM do not require notification. Notification is required if ACM/ACBM is disturbed or is found to no longer be in 'good' condition. Disturbance of ACM/ACBM is caused when any process, which through direct contact, vibration, air movement, or other means, physically moves ACM/ACBM in such a way that fibers could become airborne. If there is any doubt as to whether ACM/ACBM could be disturbed during an operation, the Asbestos Consultant should be consulted.

4.4.1.1 Planned Operations Involving ACM/ACBM

The following shall be followed depending on the amount and type of ACM/ACBM. Thermal insulation is generally measured in linear feet, surfacing material in ft.², and misc. ACM/ACBM is measured as appropriate.

- Under 3 ft.²/3 linear ft.

No notifications are required for this amount. Any ACM/ACBM waste shall be disposed of in accordance with the proper OSHA and EPA regulations.

- 3 ft.²/3 linear ft. – 160 ft.²/260 linear ft.

This requires at least a 10 working days' notice to the Illinois Department of Public Health before any ACM/ACBM is disturbed.

- Over 160 ft.²/260 linear ft.

This requires at least a 10-working-day notice to the Illinois Department of Public Health and Cook County before any ACM/ACBM is disturbed.

- Demolitions and Renovations.

For buildings being demolished, all RACM as defined in 'EPA NESHAP regulations 40CFR 61 Subpart M: 'National Emissions Standard for Asbestos'; section 141: 'Definitions' must be abated as stated in section 60.145: 'Standard for Demolition and Renovation'. Refer to the notification requirements as stated in section 60.145(b): 'Notification Requirements', for specific details. Generally, the notification timeline is 10 working days prior to the date the demolition begins, a detailed notice must be provided to the administrator. All asbestos abatement work and notifications are performed by a state-licensed asbestos abatement contractor.

4.4.1.2 Emergencies Involving ACM/ACBM

In the event of an accidental release of more than one pound of asbestos (ref. 40CFR302.4, 'List of Hazardous Substances and Reportable Quantities'). The reporting process begins with the Illinois Emergency Management agency. Notifications shall be made as early as possible. The affected area must be properly regulated in accordance with the appropriate OSHA regulations.

4.5 Warning Signs and Labels

Warning signs are placed in routine maintenance areas where ACM/ACBM is located and has a possibility of becoming disturbed during routine maintenance and/or construction. This is done to inform personnel that there is a potential hazard of asbestos fiber exposure if the ACM/ACBM is disturbed. Warning labels are placed on actual ACM/ACBM to prevent accidental damage during routine maintenance and/or construction.

4.5.1 Warning Signs

'The Asbestos Hazard Emergency Response Act of 1986' (AHERA) mandates that warning signs be displayed where there is friable and non-friable ACM/ACBM in routine maintenance areas. Warning labels are placed directly on the ACM/ACBM, such as pipe insulation, or a sign is placed in an obvious location, such as each entry location, to be seen prior to entering an area containing ACM/ACBM so those entering can take the necessary precautions before entering. Per AHERA, the warning labels shall read:

**CAUTION: ASBESTOS. HAZARDOUS.
DO NOT DISTURB WITHOUT PROPER
TRAINING AND EQUIPMENT**

Per **29CFR 1910.1001**, warning signs shall be posted at all approaches to regulated areas so that an employee may read the signs and take necessary protective steps before entering the area."

29CFR1910.1001: Occupational Safety and Health Standards, sub-part title 'Toxic and Hazardous Substances', sub-part (j)(4)(ii)(A) says that these warning signs "shall bear the following legend:"

**DANGER
ASBESTOS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
AUTHORIZED PERSONNEL ONLY**

In addition, **29CFR1926.1101**: 'Safety and Health Regulations for Construction', sub-part title 'Toxic and Hazardous Substances', sub-part (k)(7)(ii)(A) says that "The warning signs required by paragraph (k)(7) of this section shall bear the following information

**DANGER
ASBESTOS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
AUTHORIZED PERSONNEL ONLY"**

4.5.2 Warning Labels

Under **CFR1910.1001(j)(5)(i)**, warning labels shall be placed on “all raw materials, mixtures, scrap, waste, debris, and other products containing asbestos fibers, or to their containers.” Sub-part (j)(5)(ii) says that the labels will read:

**DANGER
CONTAINS ASBESTOS FIBERS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
DO NOT BREATHE DUST
AVOID CREATING DUST**

In addition, **CFR1926.1101(k)(8)(i)** states “Labels shall be affixed to all products containing asbestos and to all containers containing such products, including waste containers. Where feasible, installed asbestos products shall contain a visible label.” Per sub-part (k)(8)(iii), this label shall read:

**DANGER
CONTAINS ASBESTOS FIBERS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
DO NOT BREATHE DUST
AVOID CREATING DUST**

5.0 Locations of ACM/ACBM

Currently, ACM/ACBM is known to be present in most buildings at the Jesse Brown VAMC campus. The most recent asbestos survey was completed in 2019. The complete survey is located in the Engineering Department.

This inspection breaks down the quantity, type and location of known ACM/ACBM by building and is updated every 6 months to document any changes in the condition of the materials.

6.0 Training

Asbestos awareness training courses are available on the VA Talent Management System (TMS) and assigned by each service as required. The following training is recommended for appropriate personnel:

6.1 Asbestos Program Manager and Industrial Hygienist

No certification is required for the Asbestos Program Manager, but it is recommended that they be trained in an EPA approved Asbestos Inspector or Asbestos Management Planner course. The VA’s Professional Industrial Hygienist (VPIH) must meet the requirements of a Professional Industrial Hygienist (PIH) and may be a Certified Industrial Hygienist (CIH).

6.2 Maintenance Staff

All Maintenance and Operations (M&O) personnel that may encounter ACM/ACBM will receive 2 hours of asbestos awareness training and an annual refresher training.

6.3 Other Building Employees

All personnel that may encounter ACM/ACBM will receive 2 hours of asbestos awareness training.

6.4 Abatement Contractors

Abatement contractors are required to be state-licensed in Illinois. The contractor's Industrial Hygienist must be a PIH and may be a CIH in order to take ACM samples.

6.5 Abatement Workers

The asbestos abatement workers hired by the abatement contractors are required to be trained and state-licensed in Illinois.

7.0 Emergency Response

An asbestos fiber release emergency occurs when there is a possibility of a fiber exposure. This can happen if the ACM/ACBM is reduced from 'good' condition to 'damaged' or 'significantly damaged'. The ACM/ACBM may degrade due to age, moisture, or physical damage. Maintenance staff should always be on the lookout for any possible evidence of fiber release. Any asbestos related problems should be reported immediately to the Asbestos Program Manager, the Industrial Hygienist, the Maintenance and Operations Chief, and the FMS Chief.

7.1 Minor Fiber Release Episode

A minor fiber release episode occurs when less than 3 square feet/3 linear feet of friable ACM/ACBM becomes damaged. Examples include:

- A small section of damaged pipe from a pipe leak.
- 3 or fewer dislodged or damaged floor tiles.
- <3 sf of plaster or drywall system

This can be a result of accidental damage during maintenance or repair, or other physical damage and must be addressed immediately. The following procedures must be used to abate the asbestos by the asbestos by contractor.

7.1.1 Procedures for control:

1. Notify the Asbestos Program Manager.
2. Shut down or Isolate the HVAC system.
3. Evacuate any unauthorized personnel from the area.
4. Place signs at the entrance to the work area to prevent entrance into the area.

7.2 Major Fiber Release Episode

A major fiber release episode is a serious event and occurs when more than 3 square feet/3 linear feet of ACM/ACBM becomes damaged. An asbestos abatement contractor must be called to remove the ACM/ACBM.

7.2.1 Procedures for Control:

1. Notify the Asbestos Program Manager.
2. All unauthorized personnel will be evacuated from the room.
3. The area will be isolated as soon as possible and properly demarcated.
4. If necessary, all doors, windows and vents shall be sealed to prevent further exposure.
5. Air supply to room may need to be turned off.
6. Asbestos consultant may need to be contacted to evaluate area.
7. Abatement shall be fully documented.

8.0 Procedures for Planned Activities

8.1 Routine Maintenance in Which Disturbance to ACM/ACBM is Unlikely

These are routine maintenance procedures. The following precautions must be taken:

- The worker shall be properly trained in identifying ACM/ACBM and its' potential hazards.
- The worker shall take extra care to avoid disturbing ACM/ACBM.
- The worker shall contact the Asbestos Program Manager if ACM/ACBM becomes disturbed.

8.2 Routine Maintenance in Which Disturbance to ACM/ACBM is Likely

If disturbance to ACM/ACBM is likely, all work is to be conducted by an asbestos abatement contractor in accordance with all O&M procedures.

8.3 Quality Assurance

Air monitoring is performed during business hours for isolated areas and after regular business hours in traffic areas.

8.4 Routine Cleaning

All staff working near ACM/ACBM shall be made aware of the potential hazards before beginning work. All routine cleaning procedures performed (and restrictions) during construction are outlined in **29 CFR 1926.1101**. All abatement of ACM/ACBM is performed by a licensed asbestos contractor. However, after abatement, there may be traces of fibers in the area. Workers shall follow all methods outlined in **29 CFR 1910.1001(k) 'Housekeeping'**.

9.0 Periodic Surveillance

9.1 Physical Inspection

Per **40 CFR 763: 'Asbestos'; Subpart E: 'Asbestos-Containing Materials in Schools';** Section 92: 'Training and Periodic Surveillance'; sub-section (b): '*Periodic Surveillance*', a periodic surveillance physical inspection must be conducted every 6 months after a management plan is in effect by performing a visual inspection and documentation.

10.0 Periodic Evaluation to O&M Plan

The O&M Plan shall be updated regularly as changes occur or at each regular inspection. Every 6-months, the O&M Plan shall be updated. Prior to the evaluation, the following information shall be gathered:

1. Changes in federal and state regulations
2. All area and personal air monitoring data
3. Periodic surveillance reports
4. Emergency reports
5. ACM/ACBM drawings for each building
6. Any new asbestos abatement project files
7. Any other relevant record keeping files

11.0 Recordkeeping

All records relevant to ACM/ACBM and asbestos abatement will be located in Building 35, Engineering Offices with the Asbestos Program Manager.

11.1 General Provisions

The General guidelines for records requirements are stated in **40 CFR; Part 763: 'Asbestos'; Subpart E: 'Asbestos-Containing Materials in Schools'; Section 94: 'Recordkeeping'**. The following will be collected:

1. 6-month surveillance records
2. Asbestos Sample Results
3. Air Samples
4. Other Documents

11.2 Operations and Maintenance Activities Files

Any relevant documentation from O&M activities regarding asbestos abatement or newly discovered asbestos will be used to update the O&M plan maintained by the Asbestos Program Manager.